



A Final year project report on – Gamified learning site

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Abstract

This project presents the design and development of a gamified micro-learning platform aimed at improving the acquisition of technical micro-skills through short, structured, and interactive learning modules. Traditional learning systems often suffer from low engagement, reduced motivation, and limited adaptability to individual learning pace. To address these challenges, the proposed system integrates gamification techniques such as experience points, badges, progress tracking, and achievement-based rewards with a micro-learning content delivery approach.

The system is designed as a web-based application that supports user authentication, role-based access control, structured learning paths, and real-time progress tracking. A machine learning component is incorporated to analyze user performance and generate adaptive feedback aimed at improving learning efficiency and engagement. The platform also includes administrative features for content management and user monitoring.

From a design perspective, the system emphasizes usability, scalability, and accessibility while maintaining data security through secure authentication mechanisms and protected API communication. Ethical considerations such as user privacy, fair access to learning resources, and responsible use of behavioral data are also addressed within the system structure.

The proposed solution demonstrates how gamification and micro-learning can be combined to enhance learner engagement and skill retention in a digital environment. Overall, the system provides a structured and scalable approach to technical skill development while promoting consistent learning behavior through interactive and motivational design principles.

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Contents

Introduction.....	11
Aims and Objectives.....	11
Aim	11
Objectives	11
Problem Stated.....	12
System as a Solution.....	13
System Artefact.....	13
Scopes and Limitations	14
Scope.....	14
Limitations	14
Literature Review.....	15
Research Paper Study	15
Deterding et al. (2011) — Defining Gamification.....	15
Journal Study	16
Sailer & Homner (2020) — The Gamification of Learning: A Meta-Analysis.....	16
Cepeda et al. (2006) — Distributed Practice in Verbal Recall	17
Corbeil (2023) — Microlearning and Multimedia Design	17
Analysing Similar Sites.....	17
Codecademy (codecademy.com)	17
Duolingo (duolingo.com).....	18
Khan Academy (khanacademy.org).....	19
HackerRank (hackerrank.com)	20
LeetCode (leetcode.com)	21
Comparison to Similar Sites	22
Academic Questions	24
Question 1: Does gamification reliably improve learning outcomes, or does it only improve engagement metrics?.....	24
Question 2: What cognitive science principles best support technical micro-skill acquisition?	24
Question 3: How should payment-based access models be designed to avoid creating educational inequality?	25
System Specification.....	26

System Artefact: Feature Breakdown	26
Learner Domain Features.....	26
User Authentication	26
Quest Browsing and Difficulty Selection	26
Lesson Participation.....	26
Quiz Attempt and Feedback.....	27
Progress Tracking	27
Rewards and XP System.....	27
Payment-Based Content Unlocking.....	27
Admin Domain Features	28
Admin Authentication and Access Control	28
Quest and Lesson Management	28
User Monitoring.....	28
System Requirements Specification (SRS).....	29
Functional Requirements	29
Non-Functional Requirements	30
Design Artefact	32
Functional Decomposition Diagram (FDD)	32
Entity Relationship Diagram (ERD).....	34
User Domain ERD	34
Quest and Lesson Domain ERD	35
Progress Domain ERD.....	35
Payment Domain ERD.....	35
Reward Domain ERD	36
Use Case Diagram.....	36
Activity Diagram	39
Overall System Activity Diagram.....	39
Admin Activity Diagram	40
Sequence Diagram	41
Sequence Diagram A: Payment-Based Content Unlocking.....	41
Sequence Diagram B: Lesson Completion, Evaluation, and Rewards	42
Class Diagram.....	44
Quest Domain Classes	44
User Domain Classes	45

Supporting Domain Classes	45
Work Breakdown Structure (WBS)	45
Wireframes.....	48
Software / Project Methodology	49
Explanation of Used Methodology	49
Why Agile Was Used.....	49
Tools and Technology.....	50
Testing Approach.....	51
Explanation of Used Approach	51
Why This Approach Was Used.....	52
Test Execution Evidence.....	52
Conclusion	64
Critical Evaluation	66
Achievement Against Objectives.....	66
Challenges Encountered.....	66
Payment Integration Complexity	66
JWT Token Management Across OAuth Flows	67
Embedded Document vs. Reference Trade-offs	67
Scope Management	67
Comparison to Similar Platforms.....	67
Future Improvements	68
Evidence of Project Management	69
Log Sheet	69
Gantt Chart.....	70
Risk Management	70
Risk Register Narrative	70
References	76
Appendix.....	79
User journey summary	79
Landing page.....	79
Tutorial mode.....	79
Login/Signup page	80
User dashboard.....	81
Quest catalogue	81

Quest detail	82
User profile page.....	83
Settings.....	83
GitHub Commit History	84
PRF	86
Log sheets	90

Table of Figures

Figure 1: Codecademy dashboard.....	18
Figure 2: duolingo dashboard	19
Figure 3: Khan academy dashboard.....	20
Figure 4: Hacker rank dashboard.....	21
Figure 5: leet code.....	22
Figure 6: Functional Decomposition Diagram (FDD).....	33
Figure 7: Entity Relation Diagram (ERD)	34
Figure 8: Use case Diagram.....	38
Figure 9: Overall system Activity Diagram.....	39
Figure 10: Admin Activity Diagram.....	40
Figure 11: Sequence diagram of the payment process.....	42
Figure 12: Sequence diagram for quest completion, feedback and rewards.....	43
Figure 13: Class diagram	44
Figure 14: Sign up page wireframe	48
Figure 15: Login page wireframe	48
Figure 16: test case - user login	53
Figure 17: test case: accessing beginner free lessons using user/admin token.....	54
Figure 18: test case: accessing restricted content without payment	55
Figure 19: ids of contents purchased by the user.....	56
Figure 20: test case - the user has access to the paid quest.....	56
Figure 21: test case - lesson completion with correct asnwers	57
Figure 22: test case - lesson completion with incorrect asnwers	58
Figure 23: test case - adding a quest as admin.....	59
Figure 24: mongodb compass showing the recently added quest.....	59
Figure 25: test case: role based access control.....	60
Figure 26: test case - updating existing quest as an admin	61
Figure 27: mongodb showing the updated quest hastags section	62
Figure 28: Log sheet 1	69

Figure 29: Gantt-chart of the project	70
Figure 30: Risk register.....	70
Figure 31: Site's landing page.....	79
Figure 32: tutorial page.....	79
Figure 33: Login page.....	80
Figure 34: sign up page.....	80
Figure 35: skillquest learner dashboard	81
Figure 36: lesson player showing lesson content.....	82
Figure 37: lesson player showing the quiz.....	82
Figure 38: user profile.....	83
Figure 39: settings page	84
Figure 40: github first commit	84
Figure 41: rest of github commits	85
Figure 42: rest of github commits	85
Figure 43: rest of github commits	85

Introduction

The rapid expansion of the technology sector has created increasing demand for professionals with precise, demonstrable technical skills. While online learning platforms have proliferated in recent years, research consistently identifies low engagement, poor motivation, and weak long-term retention as persistent barriers to effective self-directed technical learning (Caballé & Clarisó, 2014). Learners often begin courses but fail to complete them, and knowledge gained is frequently lost within weeks due to the absence of structured repetition, active recall, and motivational reinforcement.

This project addresses these challenges through the development of a Gamified Micro-Learning Platform for Technical Micro-Skills, a web application that structures technical education as a series of short, focused learning quests enhanced with gamification elements including XP rewards, badges, streaks, progress tracking, and tiered content access. The platform is designed to promote sustained engagement, reward demonstrated competence, and support both beginner and advanced learners through an adaptive, role-based system.

Aims and Objectives

Aim

To design and develop a gamified micro-learning platform that enhances learner engagement and knowledge retention through structured lessons, quizzes, progress tracking, and reward mechanisms.

Objectives

- To design a role-based system supporting learners and administrators with distinct privileges and workflows.
- To implement structured learning content organised into quests and lessons with difficulty tiers (Beginner, Intermediate, Advanced).

- To integrate quiz-based assessment for knowledge evaluation and active retrieval practice.
- To track learner progress and award XP and badges based on quiz and quest performance.
- To provide administrative tools for managing learning content, users, and engagement metrics.
- To implement a payment-based unlocking system for monetisation of advanced content.
- To ensure the system is scalable, secure, and aligned with software engineering best practices.

Problem Stated

Traditional online learning platforms suffer from several well-documented weaknesses. Learners frequently experience motivational decline after initial engagement, as content delivery is passive and feedback is limited. Without structured rewards, progress signals, or adaptive challenge, completion rates on major MOOC platforms remain below 10% (Rashid et al., 2023).

Existing platforms such as Codecademy and LeetCode provide structured exercises but lack personalisation, spaced repetition scheduling, and robust gamification that aligns mechanics to learning outcomes. Many platforms over-rely on competition-based features such as global leaderboards, which benefit high-performing users while demotivating the majority.

Furthermore, technical micro-skills are narrowly scoped, repeatable competencies such as writing a correct SQL query, executing a Git rebase, or configuring environment variables are particularly susceptible to forgetting without deliberate retrieval practice. Current platforms rarely schedule revisits to specific micro-skills or integrate feedback loops that reinforce mastery at the task level.

System as a Solution

The proposed gamified micro-learning platform addresses these challenges by combining structured micro-learning content with evidence-based gamification mechanics and a flexible role-based architecture. Learners engage with content through quests composed of sequential lessons and quizzes, promoting active participation and immediate feedback.

The system introduces a tiered difficulty model (Beginner, Intermediate, Advanced) with payment-based unlocking for premium content, creating both free-to-access engagement and sustainable monetisation. Gamification elements including XP, badges, streaks, and progress dashboards are aligned to demonstrated competence rather than mere activity, supporting intrinsic motivation and mastery-oriented learning behaviours.

On the administrative side, the platform provides content management tools for creating and editing quests, lessons, and quizzes, as well as user monitoring features for tracking engagement and completion metrics. This dual-domain architecture ensures that both learner experience and platform governance are fully supported.

System Artefact

The developed artefact is a full-stack web application built on the MERN stack:

- Frontend: React 19 + Vite + Tailwind CSS + Framer Motion (hosted on a local development server)
- Backend: Node.js + Express.js REST API
- Database: MongoDB via Mongoose ODM
- Authentication: JWT tokens + Google OAuth via Passport.js
- Email: Nodemailer for account verification
- Payments: Third-party payment provider integration with webhook handling

The platform provides two distinct dashboards: a learner dashboard for content consumption and progress tracking, and an admin dashboard for content creation and user management. Key features include quest browsing, lesson participation, quiz attempt and retry, XP and badge awards, streak maintenance, and payment-based content unlocking.

Scopes and Limitations

Scope

- User authentication via email/password and Google OAuth
- Admin CRUD operations for quests, lessons, and quizzes
- Learner quest browsing, lesson participation, quiz attempt, and progress tracking
- XP, badge, and streak reward system
- Payment-based unlocking of Intermediate and Advanced content
- Role-based access control (admin and learner roles)
- RESTful API backend with MongoDB persistence

Limitations

- The gamification engine (adaptive XP scaling, dynamic difficulty adjustment) is partially implemented; full personalisation based on learner profile is planned for future development.
- Email verification service (SMTP via Nodemailer) is configured but not fully tested in a public environment.
- Load testing at production scale is outside scope; the system has been tested in a local development environment only.
- Platform-specific UI rendering differences across browsers and mobile devices have not been exhaustively tested.
- The content library currently contains seed data; a full microskill curriculum (Git, SQL, JavaScript, Docker) is planned but not yet authored.

Literature Review

The literature underpinning this project spans four intersecting domains: gamification mechanics and their pedagogical effects; adaptive and personalised gamification; microlearning theory and cognitive science principles of spaced and retrieval practice; and empirical evidence from existing technical learning platforms. Together these bodies of research motivate the design decisions made in the developed platform and identify gaps that the system attempts to address.

The application of game-design elements in non-game contexts has been proposed as a powerful mechanism for increasing learner engagement and promoting sustained practice (Deterding et al., 2011). However, meta-analytic evidence shows that gamification reliably increases engagement metrics such as time-on-task but produces heterogeneous effects on learning outcomes, with quality depending heavily on how mechanics are aligned to instructional goals (Hamari, Koivisto & Sarsa, 2014; Sailer & Homner, 2020). Cognitive science research additionally establishes spaced practice and retrieval-based testing as the most robust techniques for long-term retention (Roediger & Karpicke, 2006; Cepeda et al., 2006), and these principles map naturally onto microlearning's short, repeated practice sessions.

Research Paper Study

Deterding et al. (2011) — Defining Gamification

Deterding, Dixon, Khaled and Nacke's foundational 2011 paper provided the first widely adopted taxonomy of gamification elements, distinguishing between game mechanics (points, levels, badges), game dynamics (challenge, competition, cooperation), and game aesthetics (emotion, immersion). This framework informed the selection of gamification elements in the platform, specifically, the choice of mastery-oriented badges, XP-based progression, and streak maintenance as mechanisms that align to competence demonstration rather than superficial activity.

Hamari, Koivisto & Sarsa (2014) — Does Gamification Work?

This systematic review of 24 empirical studies on gamification confirmed consistent positive effects on engagement and participation but noted highly context-dependent effects on learning outcomes. Critically, the review identified that gamification tends to work better when elements require demonstrated competence (skill-check quizzes, mastery badges) rather than rewarding uninformed activity (points for every click). This finding directly informed the design of the quiz-gated XP system in which learners must correctly answer quiz questions to progress and earn rewards.

Roediger & Karpicke (2006) — Test-Enhanced Learning

This landmark study demonstrated that retrieval practice and testing oneself on learned material produces dramatically better long-term retention than restudying the same material. The testing effect provides strong evidence for embedding short quizzes at the end of every lesson rather than treating them as optional assessments. The platform's mandatory quiz structure before lesson completion directly implements this principle.

Rapp et al. (2022) — Tailored Gamification in Education

Rapp and colleagues' literature review identified that one-size-fits-all gamification consistently produces uneven outcomes across learner populations. Their recommendation for adaptive gamification is adjusting which elements are visible based on learner profile and motivation orientation. It motivated the platform's future roadmap for personalisation features, even where full adaptive gamification was outside the current scope.

Journal Study

Sailer & Homner (2020) — The Gamification of Learning: A Meta-Analysis

Published in Educational Psychology Review, this meta-analysis of 30 studies found that gamification had a medium-to-large effect on cognitive learning outcomes when gamification

elements were meaningfully integrated with learning content, but near-zero effects when mechanics were superficially layered onto existing instructional designs. The study emphasised that badge systems linked to demonstrated skill mastery were consistently more effective than points-for-participation systems, reinforcing the mastery-badge design used in this platform.

Cepeda et al. (2006) — Distributed Practice in Verbal Recall

This large-scale quantitative synthesis published in *Psychological Bulletin* confirmed the spacing effect across hundreds of studies: distributing practice over time dramatically outperforms massed study for long-term retention. For micro-skill platforms, this supports scheduling revisits to content and structuring lessons as short, revisitable modules rather than single extended sessions. A design principle reflected in the platform's quest-and-lesson microlearning architecture.

Corbeil (2023) — Microlearning and Multimedia Design

This study demonstrated that microlearning combined with multimedia design significantly improved knowledge acquisition and skill retention compared to traditional e-learning formats. Short, focused learning units with immediate feedback loops were identified as particularly effective for procedural skill acquisition directly supporting the platform's lesson-quiz-feedback cycle for technical micro-skills.

Analysing Similar Sites

Codecademy (codecademy.com)

Codecademy is a browser-based interactive coding platform offering structured courses in programming languages, data science, and web development. Lessons are presented as short modules with in-browser coding exercises and immediate feedback. The platform uses a course-completion progress bar and achievement badges, but gamification is relatively shallow. Badges are awarded for course completion rather than demonstrated competence on individual skills. The platform lacks spaced repetition scheduling and has limited adaptive personalisation. Pricing is subscription-based with a free tier offering limited content.

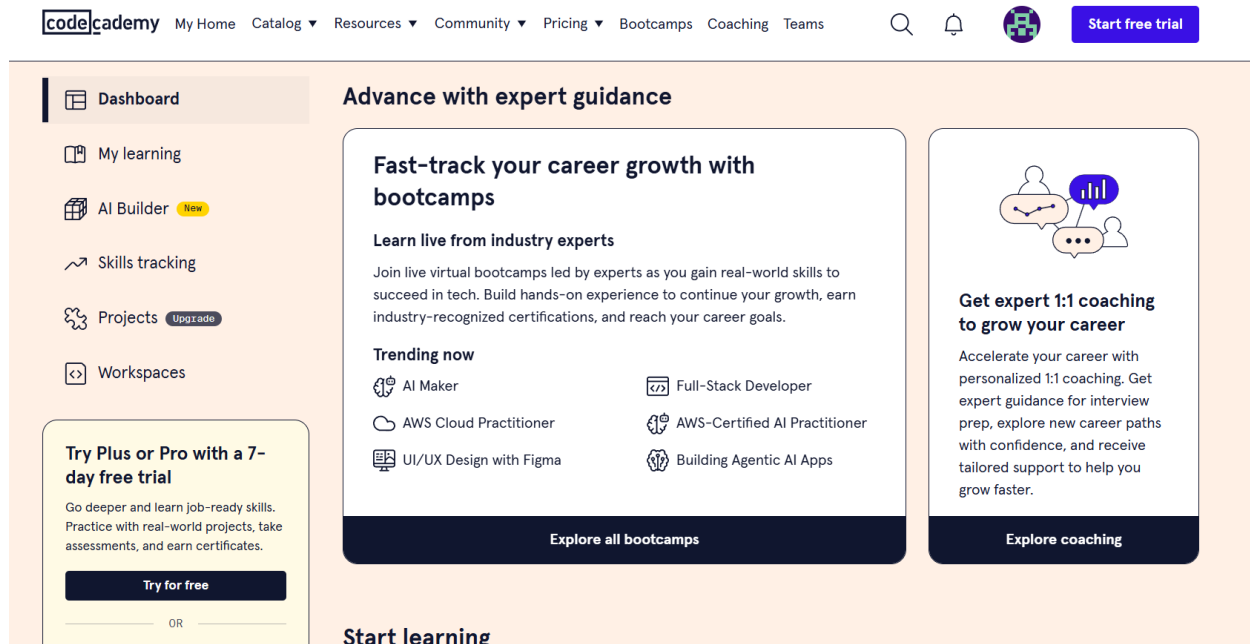


Figure 1: Codecademy dashboard

Duolingo (duolingo.com)

Duolingo is the most widely used gamified language-learning application, featuring XP systems, streak tracking, leaderboards, hearts (lives), and gem currencies. Its gamification design is highly sophisticated and has been studied extensively as a model of engagement-oriented design. Crucially, Duolingo's lessons are structured as short, daily micro-practice sessions that are directly aligned with spacing and retrieval principles. However, Duolingo's content is limited to language learning and its competitive features (leaderboards) have been criticised for demotivating non-competitive learners.

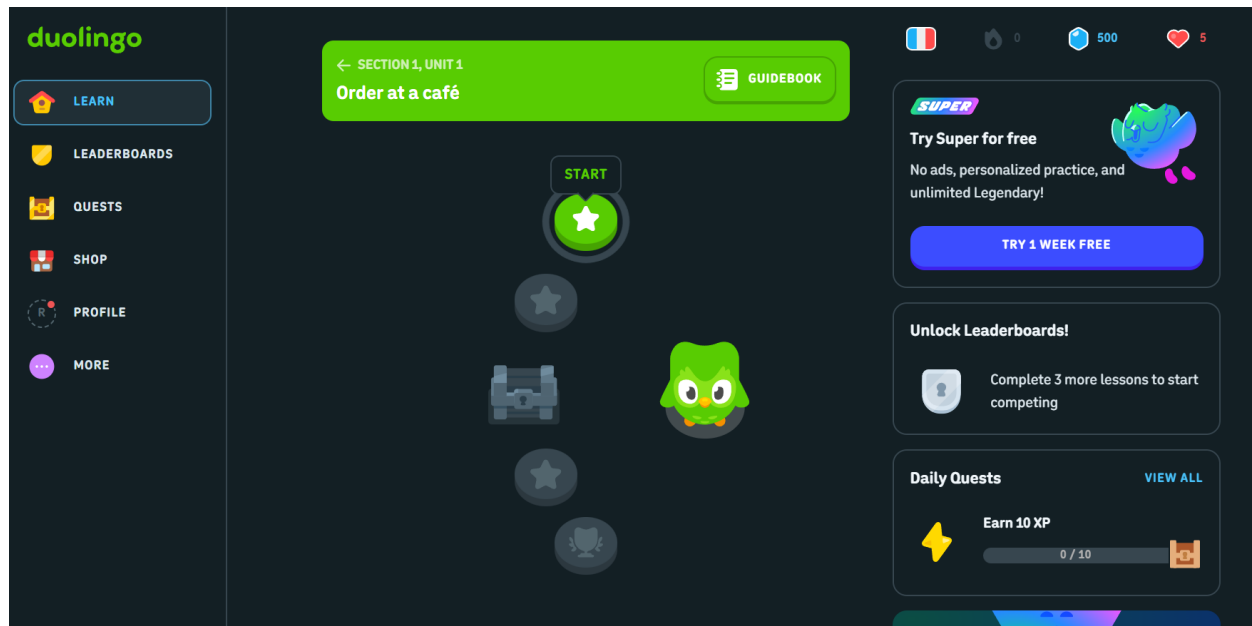


Figure 2: duolingo dashboard

Khan Academy (khanacademy.org)

Khan Academy provides free educational content across mathematics, sciences, computing, and humanities. Its gamification includes mastery points, badges, and a skill mastery system that requires repeated correct answers before advancing. The platform's mastery model is closely aligned with evidence-based learning: learners must demonstrate competence before progressing, and content is revisited if performance declines. However, the platform's gamification is conservative and may not provide sufficient motivational pull for self-directed adult learners seeking technical micro-skills.

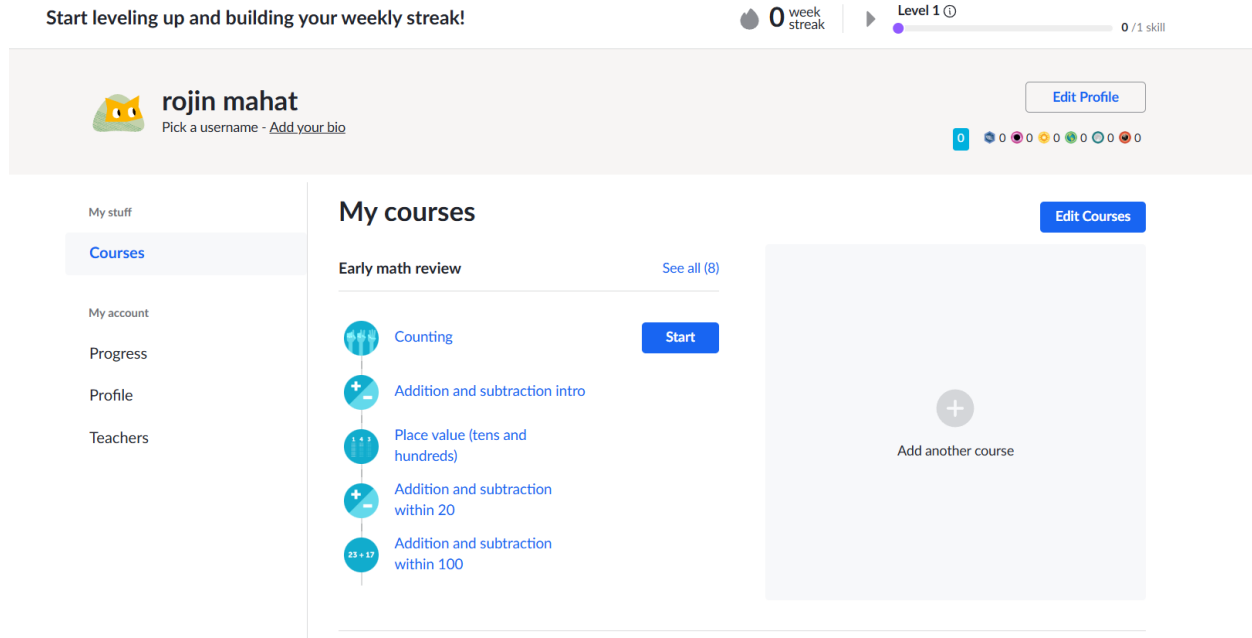


Figure 3: Khan academy dashboard

HackerRank (hackerrank.com)

HackerRank is a skills-based assessment platform used primarily by companies for developer hiring. It provides coding challenges, certificates, and skill badges across multiple programming domains. The platform excels at technical rigour but is competition and assessment-oriented rather than learning-oriented. Feedback is limited to pass/fail on challenge submissions, and the absence of structured pedagogical scaffolding makes it unsuitable as a primary learning environment for beginners.

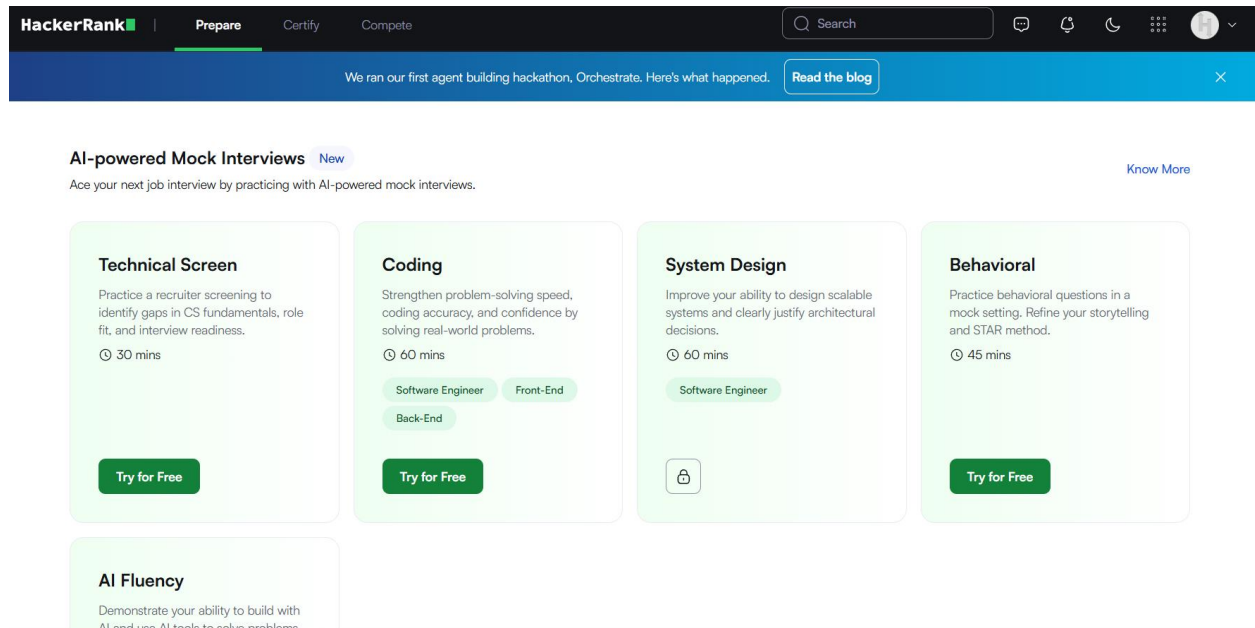


Figure 4: Hacker rank dashboard

LeetCode (leetcode.com)

LeetCode is a popular platform for competitive programming practice, offering thousands of algorithmic challenges across difficulty tiers. It features a points system, contest leaderboards, and premium subscription content. Like HackerRank, LeetCode is assessment-heavy and community-driven, making it effective for advanced practitioners but poorly suited for structured micro-skill acquisition by beginners. The absence of instructional content means learners must bring prior knowledge before the platform adds value.

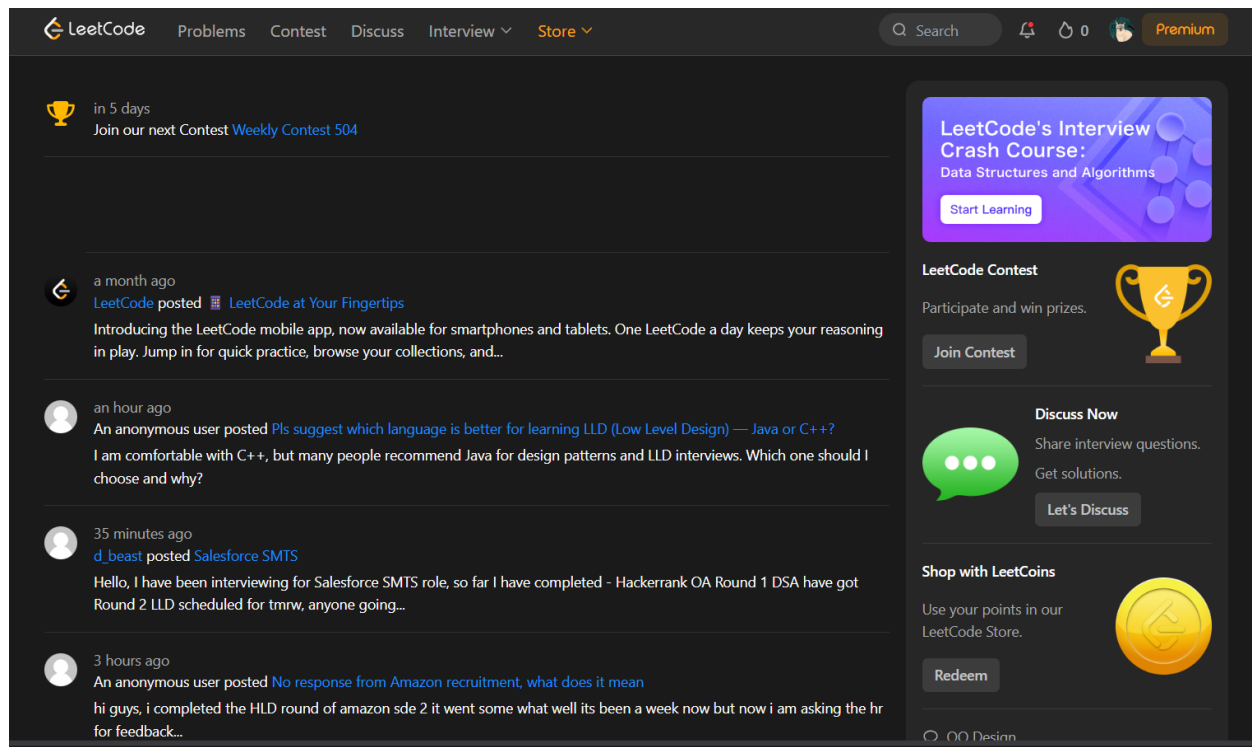


Figure 5: leet code

Comparison to Similar Sites

Feature / Criterion	This Platform	Codecademy	Duolingo	Khan Academy	HackerRank
Gamification Depth	High (XP, badges, streaks, levels, rewards)	Medium (badges, progress bars)	Very High (XP, streaks, leaderboards, hearts)	Medium (mastery points, badges)	Low (certificates, rank)
Mastery-Aligned Rewards	Yes — XP gated by quiz correctness	Partial — completion-based	Yes — repetition required	Yes — mastery model	No — challenge-based
Microlearning Architecture	Yes — short quests & lessons	Yes — short modules	Yes — 5–10 min sessions	Yes — short videos & exercises	No — open-ended challenges

Feature / Criterion	This Platform	Codecademy	Duolingo	Khan Academy	HackerRank
Spaced/Retrieval Practice	Planned (partial)	No	Yes (spaced repetition)	Yes (mastery retesting)	No
Admin Content Management	Yes — full CRUD admin dashboard	No (platform-managed)	No (platform-managed)	No (platform-managed)	No (platform-managed)
Role-Based Access	Yes (admin / learner)	No	No	No	Limited (coach / student)
Payment-Based Unlocking	Yes (Intermediate/Advanced tiers)	Yes (Pro subscription)	Yes (Super Duolingo)	No (fully free)	Yes (premium challenges)
Free Foundational Content	Yes (Beginner tier free)	Limited	Yes	Yes (all content free)	Limited
Technical Micro-Skill Focus	Yes (Git, SQL, JS, Docker planned)	Yes (programming languages)	No (language learning)	Partial (computing section)	Yes (algorithms, DS)
Personalisation / Adaptive	Planned	Limited	Yes (adaptive practice)	Yes (mastery-based pacing)	No

Academic Questions

Question 1: Does gamification reliably improve learning outcomes, or does it only improve engagement metrics?

The research suggests that the relationship between gamification and learning outcomes is nuanced and context-dependent. Meta-analytic evidence (Hamari et al., 2014; Sailer & Homner, 2020) confirms that gamification consistently improves engagement metrics like time-on-task, participation rates, and completion, but produces heterogeneous effects on actual learning outcomes. The critical determining factor is the alignment between gamification mechanics and instructional goals. When badges and points require demonstration of task-specific competence (e.g., correctly answering a mastery quiz before earning XP), learning outcomes improve significantly. When mechanics merely reward activity regardless of quality (points for any click, badges for logging in), learning outcomes are not reliably improved and may actually decline as learners optimise for rewards rather than understanding. This project implements the former approach: XP and badges are awarded only upon successful quiz completion, ensuring that gamification reinforces rather than replaces genuine learning.

Question 2: What cognitive science principles best support technical micro-skill acquisition?

The strongest empirical support for technical micro-skill retention comes from two related cognitive science principles: the spacing effect and the testing effect. The spacing effect (Cepeda et al., 2006) demonstrates that distributing practice over time, returning to the same material days or weeks later produces far superior long-term retention compared to massed practice in a single session. The testing effect (Roediger & Karpicke, 2006) shows that actively retrieving information from memory through low-stakes testing is more effective for long-term retention than restudying the material passively. Both effects are particularly well-suited to technical micro-skills, which are procedural and benefit from repeated correct execution across multiple sessions. The platform's quest-and-lesson structure with mandatory end-of-lesson quizzes directly implements retrieval practice, and the microlearning format (short, focused sessions) supports spaced practice when learners return on consecutive days for streak maintenance.

Question 3: How should payment-based access models be designed to avoid creating educational inequality?

The ethical tension between commercial sustainability and educational equity is a genuine challenge for freemium learning platforms. Research on digital inequality (Selwyn, 2016) and platform ethics (Floridi et al., 2018) identifies two-tier access structures as potential vectors for reinforcing existing socioeconomic inequalities: users who cannot afford premium tiers are systematically excluded from advanced educational content, compounding rather than reducing skill gaps. The design response to this tension involves several principles. First, all foundational content (Beginner tier) should remain freely accessible, ensuring that the platform delivers genuine educational value to all users regardless of financial capacity. Second, premium tiers should extend rather than restrict the core learning experience by providing additional advanced content rather than removing features from the free tier. Third, transparent pricing and fair refund policies ensure that paying users understand exactly what they are purchasing. The platform implements these principles by making Beginner-level quests and lessons freely available while requiring payment only for Intermediate and Advanced tiers, which provide enriched content rather than gating essential functionality.

System Specification

System Artefact: Feature Breakdown

Learner Domain Features

The learner domain is designed for content consumption, knowledge assessment, and personal progress tracking.

User Authentication

Learners can register using an email address and password or authenticate via Google OAuth. Email verification is supported through Nodemailer. JWT tokens are issued upon successful authentication and stored client-side to maintain session state. Google OAuth integration uses Passport.js with the GoogleStrategy.

Quest Browsing and Difficulty Selection

Authenticated learners can browse the full quest catalogue, which displays quests organised by difficulty level: Beginner, Intermediate, and Advanced. Each quest card displays the quest title, description, difficulty badge, total XP available, and reward badge. Intermediate and Advanced quests are visually marked as locked for users who have not completed the payment flow for those tiers.

Lesson Participation

Each quest contains an ordered sequence of embedded lessons. Learners progress through lessons sequentially; the system prevents skipping lessons within a quest to preserve the intended learning flow. Each lesson displays instructional content and is accompanied by a quiz.

Quiz Attempt and Feedback

After consuming lesson content, learners attempt a multiple-choice quiz associated with that lesson. Each quiz presents a question with four answer options. The system provides immediate feedback upon answer submission. Incorrect answers display the correct answer and offer the learner the option to retry the lesson before reattempting the quiz. Correct answers trigger the progress and reward update flow.

Progress Tracking

A personalised dashboard displays the learner's current XP total, level, streak count, completed quests, earned badges, and recent activity. The dashboard also surfaces recommended quests based on the learner's progress history.

Rewards and XP System

Learners earn XP upon successful quiz completion for each lesson. Upon completing all lessons in a quest, additional XP and a reward badge are awarded. The XP total is used to calculate the learner's current level. Streak counts are incremented for consecutive days of activity.

Payment-Based Content Unlocking

Learners wishing to access Intermediate or Advanced content are directed to a payment flow. The frontend initiates a payment request, the backend creates a payment intent via the third-party provider, and a webhook confirms the transaction. Upon successful payment, the locked difficulty tier is permanently unlocked for that user.

Admin Domain Features

The admin domain provides system governance, content management, and learner monitoring capabilities.

Admin Authentication and Access Control

Administrators authenticate through the same login endpoint as learners but receive a JWT token encoding the admin role. All administrative API routes are protected by an admin middleware that verifies both the JWT token and the admin role claim before permitting access.

Quest and Lesson Management

Admins can create, edit, update, and delete quests through the admin dashboard. Quest creation accepts a title, description, difficulty level, total XP, reward badge, and an array of embedded lessons. Each lesson contains a title, instructional content, XP reward, and an associated quiz with a question, four answer options, and a correct answer index.

User Monitoring

The admin dashboard surfaces learner engagement data including quest completion statistics, user registration counts, and progress records. This supports content quality assessment and identification of lessons with high failure or retry rates.

System Requirements Specification (SRS)

Functional Requirements

Req. ID	Component	Requirement Description	Priority
FR-01	Authentication	The system shall allow learners to register with a valid email and password.	High
FR-02	Authentication	The system shall authenticate users via Google OAuth and issue a JWT.	High
FR-03	Authentication	The system shall verify email addresses via a tokenized email link.	Medium
FR-04	Quest Management	The system shall display all available quests with title, difficulty, and XP.	High
FR-05	Quest Management	The system shall prevent access to difficulty tiers locked without payment.	High
FR-06	Lesson	The system shall deliver lesson content sequentially within a quest.	High
FR-07	Quiz	The system shall present a multiple-choice quiz after each lesson.	High
FR-08	Quiz	The system shall provide immediate feedback on quiz answer submission.	High
FR-09	Quiz	The system shall allow lesson retry upon incorrect quiz answer.	Medium
FR-10	Rewards	The system shall award XP upon successful lesson quiz completion.	High
FR-11	Rewards	The system shall award a badge upon full quest completion.	High

Req. ID	Component	Requirement Description	Priority
FR-12	Progress	The system shall persist and display learner progress on the dashboard.	High
FR-13	Streaks	The system shall track and display daily activity streaks.	Medium
FR-14	Payment	The system shall initiate a payment flow for locked content tiers.	High
FR-15	Payment	The system shall process payment webhooks and update user access rights.	High
FR-16	Admin	The system shall allow admins to create, edit, and delete quests and lessons.	High
FR-17	Admin	The system shall restrict admin endpoints to users with the admin role.	High
FR-18	Admin	The system shall display learner progress and engagement metrics to admins.	Medium

Non-Functional Requirements

Req. ID	Category	Requirement Description
NFR-01	Security	All API endpoints must be protected with JWT authentication.
NFR-02	Security	Passwords must be hashed before storage using bcrypt.
NFR-03	Security	Admin endpoints must enforce role-based access control.

Req. ID	Category	Requirement Description
NFR-04	Performance	API responses must complete within 500ms under normal load.
NFR-05	Usability	The UI must be responsive across desktop and mobile screen sizes.
NFR-06	Reliability	The system must handle invalid JWT tokens gracefully with 401 responses.
NFR-07	Maintainability	All code must be written in JavaScript/JSX without TypeScript.
NFR-08	Scalability	The database schema must support addition of new quest domains without migration.
NFR-09	Compliance	Payment card data must not be stored locally; a third-party provider must be used.
NFR-10	Accessibility	Interface must meet WCAG 2.1 AA contrast and navigation requirements.

Design Artefact

This chapter presents the complete set of system design diagrams produced during the design phase of the project. Each diagram targets a distinct architectural concern and together they provide a comprehensive view of system architecture, data relationships, user interactions, dynamic behaviour, and screen layout. All diagrams were developed iteratively in parallel with system implementation.

Functional Decomposition Diagram (FDD)

The Functional Decomposition Diagram illustrates the hierarchical breakdown of the system into its major functional modules and sub-functions. The root node represents the Gamified Learning Platform as a whole, which decomposes into five primary modules:

- **User Management:** covers registration, login/logout, profile management, and admin/instructor management sub-functions.
- **Lesson Management:** covers browsing the microskill catalogue, difficulty selection, accessing lesson content, and tracking lesson completion.
- **Assessment and Feedback:** covers quiz/exercise attempt, answer submission, instant feedback delivery, and retry options.
- **Gamification and Rewards:** covers the points system, badges and levels, streak maintenance, and the project/progress dashboard.
- **Payment and Monetisation:** covers transaction handling and payment-based access to Intermediate and Advanced content tiers.

Additionally, cross-cutting concerns are captured in two supporting modules:

- **Progress Tracking:** covers storing learner activity, updating progress data, and generating performance summaries.
- **Administration and System Control:** covers adding, updating, and removing lessons; monitoring learner statistics; and setting lesson difficulty levels.

The FDD establishes the modular structure of the system, ensuring separation of concerns and providing a clear mapping between user-facing features and backend service responsibilities. Each leaf node in the FDD corresponds to one or more API endpoints or frontend components in the implementation.

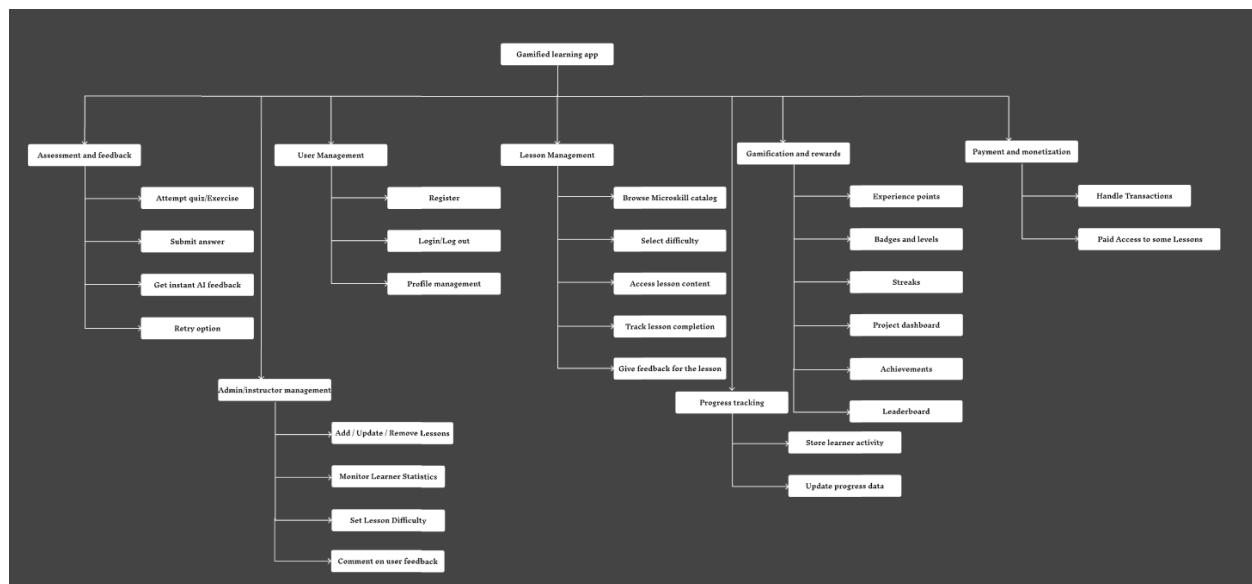


Figure 6: Functional Decomposition Diagram (FDD)

Note: The FDD diagram image is captured in the submitted Artefact Design document (Figure 3.2.1). The diagram was produced using a dedicated diagramming tool and is referenced here by its figure number to avoid image quality degradation in reproduction.

Entity Relationship Diagram (ERD)

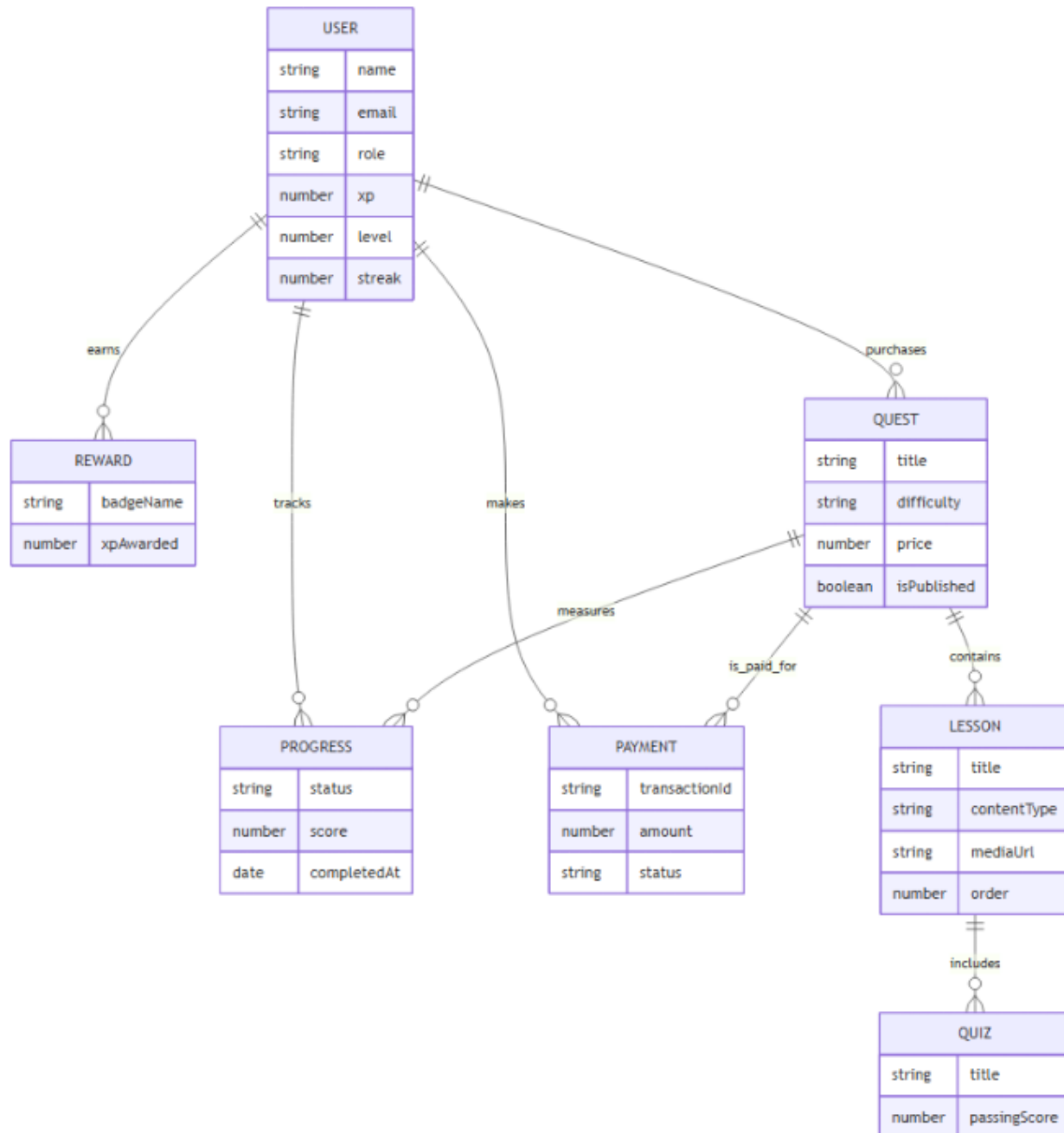


Figure 7: Entity Relation Diagram (ERD)

User Domain ERD

The User entity is central to the data model. It stores identity information (name, email, passwordHash, googleId, picture), role classification (role: learner/admin), gamification state (points, badges, level, xpToNextLevel, currentXP, streak, questsCompleted, totalQuests),

verification status (isVerified, verificationToken, verificationTokenExpires), and embedded sub-documents for recentActivity and recommendedQuests.

The User is associated with the Progress and Payment entities through one-to-many relationships: one user can have many progress records (one per lesson attempted) and many payment records (one per unlocked difficulty tier).

Quest and Lesson Domain ERD

The Quest entity stores quest metadata (title, description, difficulty, totalXP, rewardBadge, createdAt, updatedAt) and embeds an array of EmbeddedLesson sub-documents. Each EmbeddedLesson contains a title, content, order index, XP reward, and an embedded EmbeddedQuiz with a question, options array, and correctAnswer index.

A standalone Lesson entity exists for cases where lessons are referenced directly (e.g., via payment records). The standalone Lesson contains title, description, content, difficulty, tags, and an embedded Quiz array.

Progress Domain ERD

The Progress entity acts as a junction between User and Lesson, recording the completion status (in-progress / completed), score, attempt count, last attempt date, and timestamps for each user-lesson pair.

Payment Domain ERD

The Payment entity records each payment transaction, associating a userId and lessonId with the amount, currency, status (pending/completed/failed), transactionDate, transactionId, and the unlockedDifficulty tier. This allows the system to efficiently query whether a user has unlocked a given difficulty level.

Reward Domain ERD

The Reward entity stores named reward badges with a name, pointsRequired threshold, description, icon reference, and timestamps. Rewards are referenced by name in Quest and User documents.

The full ERD diagram is presented in Figure 3.4.1 of the submitted Artefact Design document. It illustrates all entity attributes, relationship cardinalities, and key embedded sub-structures.

Use Case Diagram

The Overall Use Case Diagram identifies two primary actors: Learner and Admin/Instructor and a System actor for automated processes. It maps them to the use cases available within the Gamified Learning Platform boundary.

Learner Use Cases:

- **Register:** create a new account with email or Google OAuth
- **Login:** authenticate and receive a session token
- **Browse Microskill Catalogue:** view available quests and their metadata
- **Select Difficulty (Beginner / Intermediate / Advanced):** extended by Make Payment to unlock for locked tiers
- **Attempt Lesson/Quiz:** includes Receive Immediate Feedback
- **Earn Rewards:** includes View Dashboard/Progress Tracking
- **Retry Lessons:** extended by Select Difficulty when returning to a lesson
- **Make Payment:** to unlock Intermediate or Advanced content

Admin/Instructor Use Cases:

- **Upload New Lessons:** create quest and lesson content
- **Edit/Manage Content:** modify or delete existing quests and lessons; includes Set Lesson Difficulty
- **View Learner Progress Reports:** includes View Dashboard data for all users
- **Evaluate Answers:** system-level evaluation of quiz submissions
- **Update Progress Data:** system automation following lesson/quiz completion
- **Restrict/Unlock Access to Locked Difficulty:** system response to payment events
- **Award Rewards:** system-level badge and XP distribution

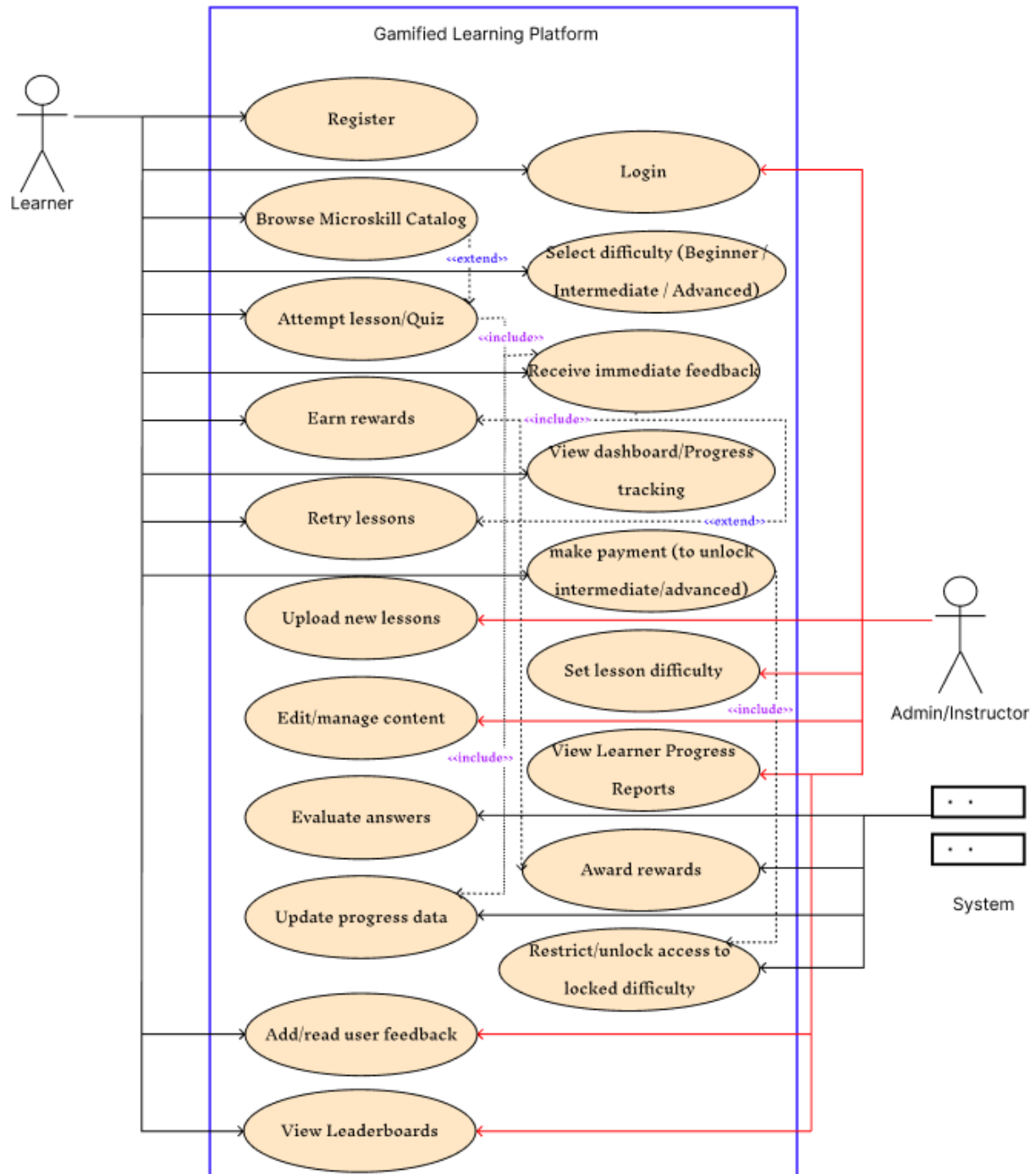


Figure 8: Use case Diagram

The Use Case Diagram confirms system boundary coverage and supports traceability from requirements to implementation.

Activity Diagram

Overall System Activity Diagram

The Overall System Activity Diagram uses swimlanes to separate Learner and System responsibilities across the end-to-end learning session workflow:

1. The learner logs in; the system verifies credentials and loads the dashboard.
2. The learner browses the quest catalogue and selects a quest.
3. The learner selects a difficulty level. If the difficulty is Beginner, the system grants access directly. If Intermediate or Advanced, the system checks payment status.
4. If payment is not confirmed, the system denies access and presents the payment flow. If payment succeeds, the system records the Payment record and grants access.
5. The learner enters the lesson and consumes the instructional content.
6. The learner attempts the quiz. The system evaluates the submission.
7. If the answer is incorrect, the system provides feedback and offers a retry option. If the learner declines to retry, the workflow returns to the dashboard. If the learner retries, the flow returns to step 5.
8. If the answer is correct, the system awards XP and updates the Progress record. If the quest is fully completed, a badge is awarded.
9. The system redirects the learner to the updated dashboard.

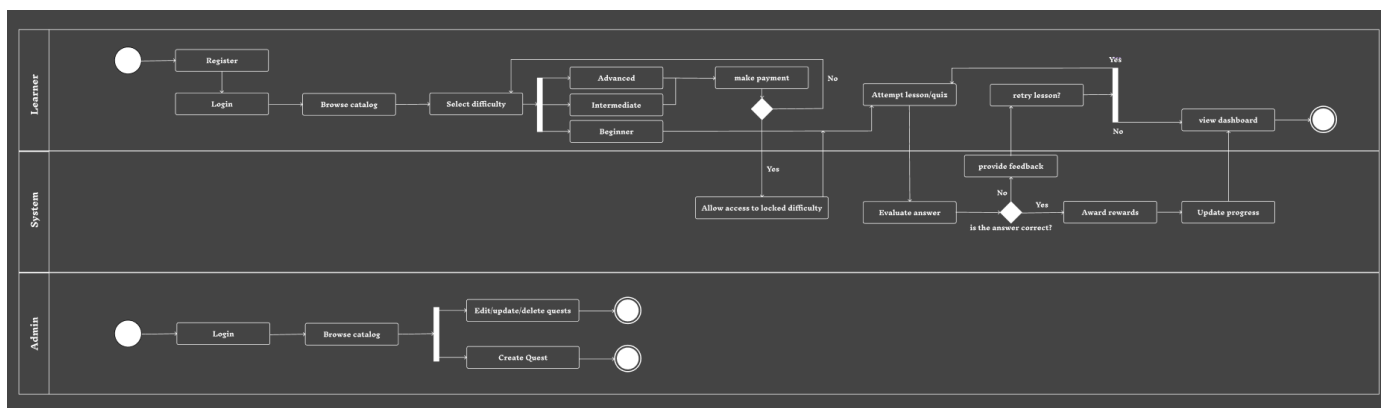


Figure 9: Overall system Activity Diagram

Admin Activity Diagram

A separate Admin Activity Diagram describes the content management workflow:

10. Admin authenticates and accesses the admin dashboard.
11. Admin selects Create Quest and provides quest metadata.
12. Admin adds one or more lessons to the quest, including lesson content, XP reward, and quiz details.
13. Admin saves the quest; the system validates and persists the new quest to MongoDB.
14. Admin can subsequently edit or delete quests via the management interface.
15. Admin views learner progress statistics and quest completion metrics.

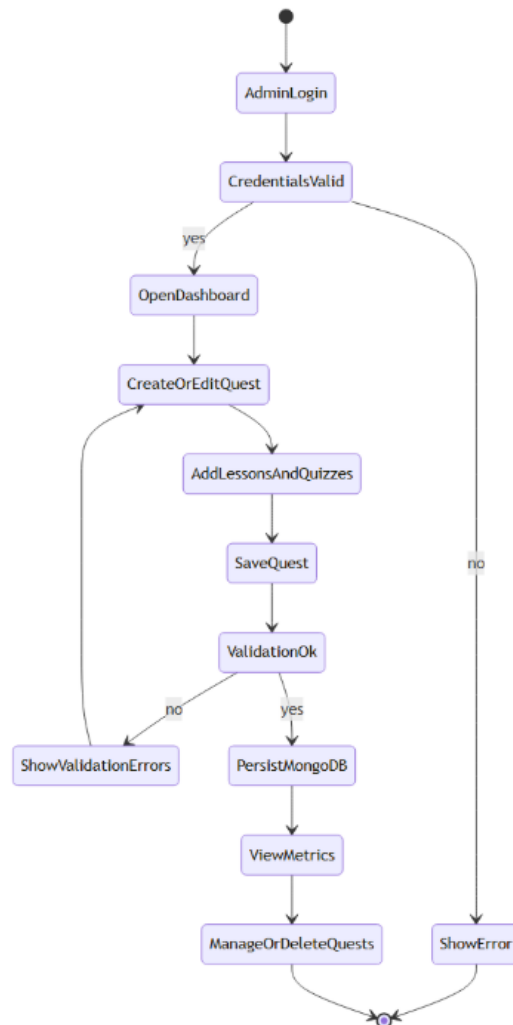


Figure 10: Admin Activity Diagram

Sequence Diagram

Sequence Diagram A: Payment-Based Content Unlocking

Actors: Learner, React App (Frontend), Backend API (Express), MongoDB, Payment Provider.

The sequence begins with the learner clicking Unlock Intermediate on a lesson card. The frontend sends a GET request to `/api/access?lessonId={L}&difficulty=intermediate` with the JWT. The backend queries MongoDB for an existing Payment record for that user-lesson-difficulty combination. If the combination is already unlocked, the backend returns 200 AccessGranted and the frontend opens the lesson content. If not unlocked, the frontend sends a POST to `/api/payments/create` with the lessonId, difficulty, and amount. The backend creates a pending Payment record in MongoDB, generates a checkout session with the Payment Provider, and returns the checkoutUrl and paymentId to the frontend. The frontend redirects the learner to the checkout page. Upon successful payment, the Payment Provider triggers a webhook to the backend with the paymentId and transactionId. The backend updates the Payment record (status: completed, unlockedDifficulty: intermediate), upserts a Progress record for the user-lesson pair, and returns a 200 confirmation. The frontend then re-queries `/api/access` and receives AccessGranted, opening the lesson.

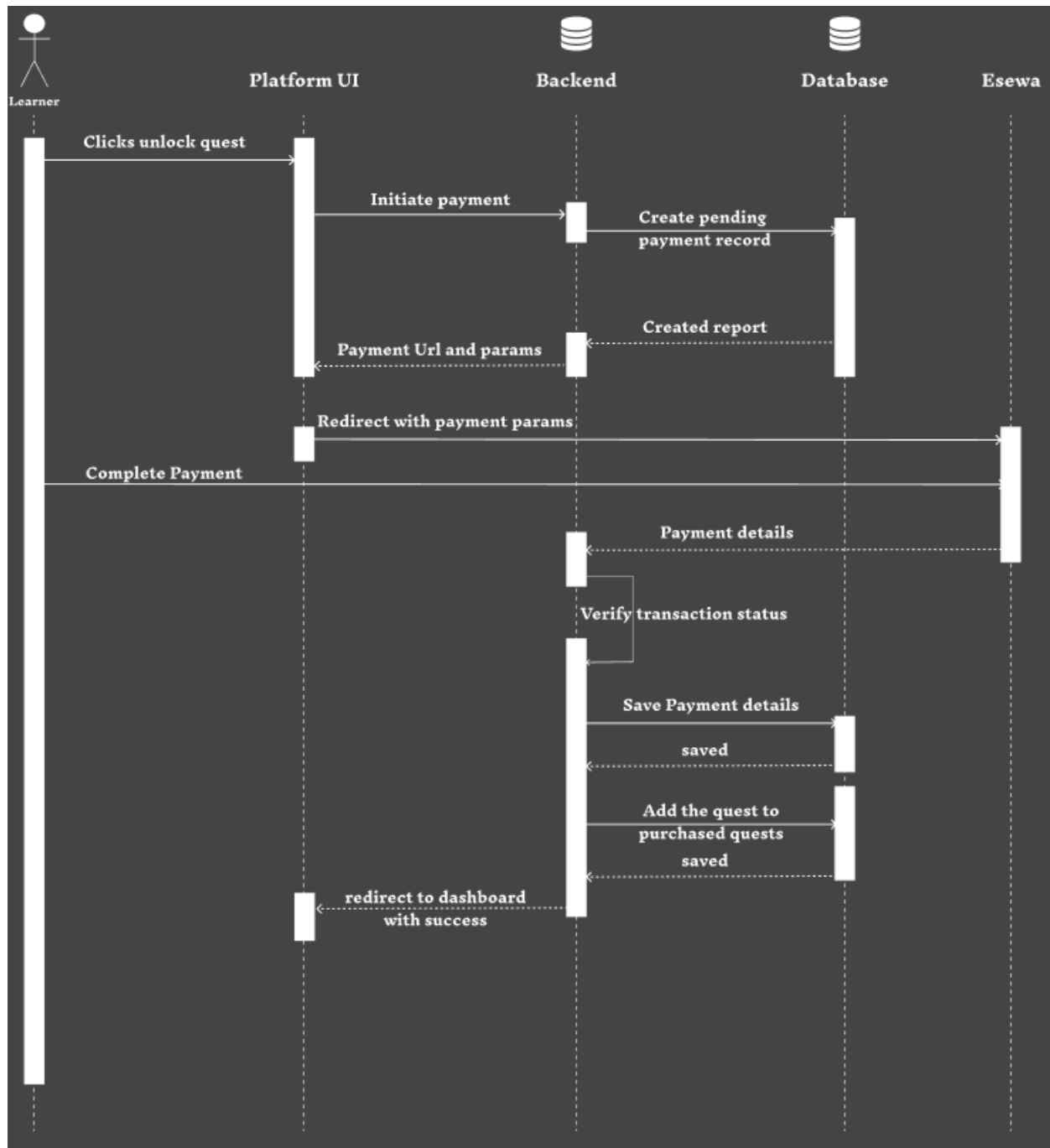


Figure 11: Sequence diagram of the payment process

Sequence Diagram B: Lesson Completion, Evaluation, and Rewards

Actors: Learner, Platform UI, Lesson Module, Evaluation Engine, Database, Reward System.

After login and authentication, the learner selects a difficulty and lesson. The Platform UI fetches lesson content from the Lesson Module, which retrieves data from the Database and returns it. The learner submits a quiz answer; the Platform UI sends it to the Evaluation Engine, which fetches the correct answer from the Database. If the answer is incorrect, the system returns to the Platform UI which shows feedback and a retry option. If the answer is correct, the Platform UI triggers the Reward System, which updates learner rewards in the Database and returns the updated reward info. The Platform UI then displays the rewards and updated progress summary.

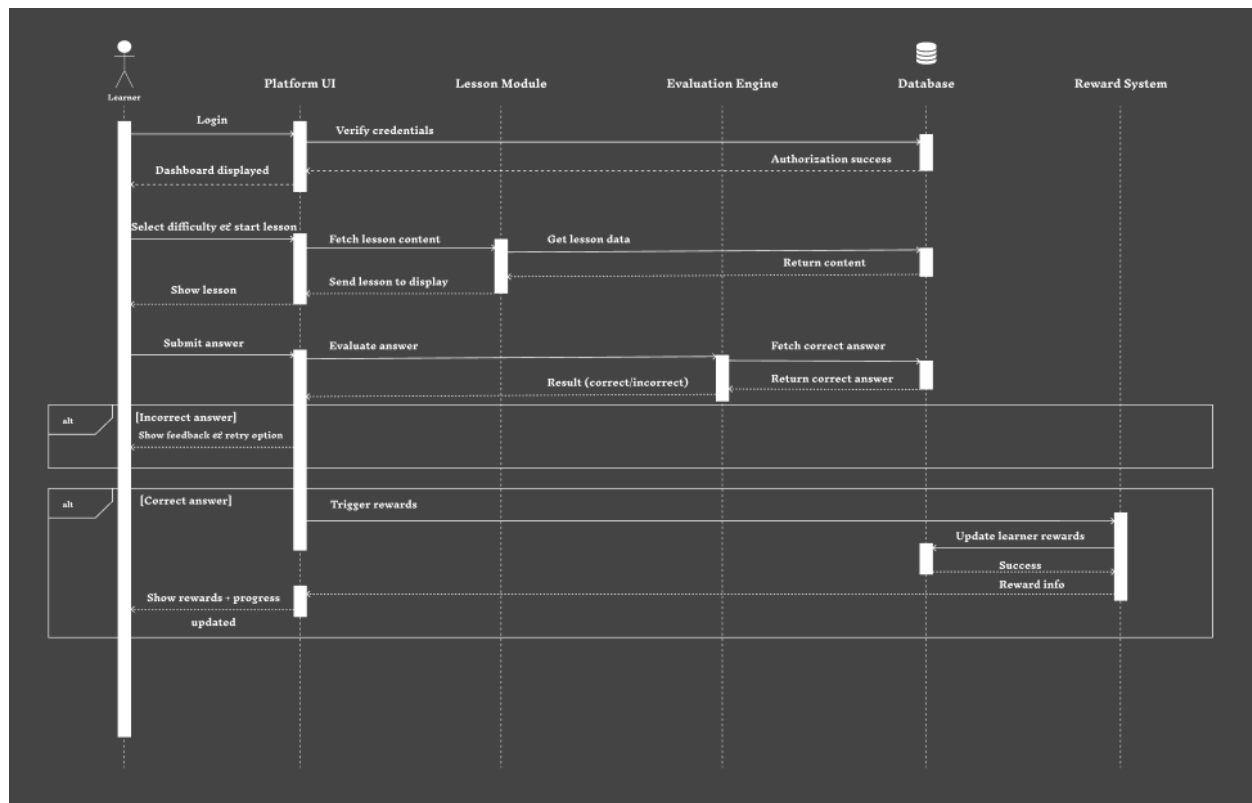


Figure 12: Sequence diagram for quest completion, feedback and rewards

Class Diagram

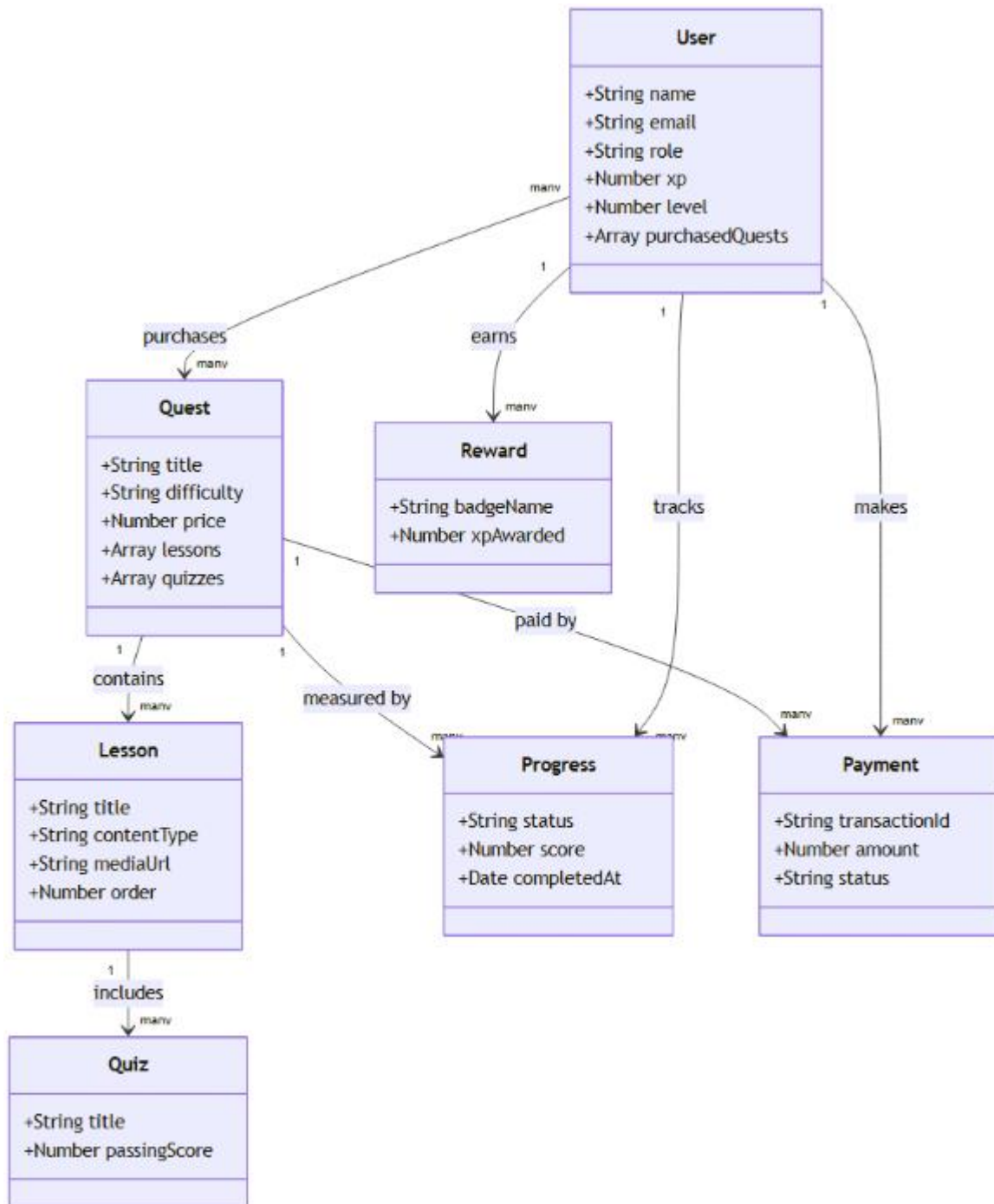


Figure 13: Class diagram

Quest Domain Classes

Quest is the primary domain class, containing attributes: `_id` (ObjectId), `title` (string), `description` (string), `difficulty` (string), `totalXP` (number), `rewardBadge` (string), `createdAt` (date), `updatedAt`

(date). Quest has a composition relationship with EmbeddedLesson: each Quest embeds an array of EmbeddedLesson objects. EmbeddedLesson contains: title (string), content (string), order (number), xp (number), quizzes (array). EmbeddedLesson has a composition relationship with EmbeddedQuiz: question (string), options (array), correctAnswer (number).

User Domain Classes

User is the central class with attributes: `_id` (ObjectId), name (string), email (string), passwordHash (string), role (string), points (number), badges (array), googleId (string), picture (string), level (number), xpToNextLevel (number), currentXP (number), streak (number), questsCompleted (number), totalQuests (number), recentActivity (array), recommendedQuests (array), skills (json), isVerified (boolean), verificationToken (string), verificationTokenExpires (date), createdAt (date), updatedAt (date). User has associations: User → Activity (recentActivity composition), User → RecommendedQuest (recommendedQuests composition).

Supporting Domain Classes

The domain model includes the following primary entities. Progress links a User to a Lesson and stores identifiers and results : `_id`, `userId`, `lessonId`, status (string), score (number), attempts (number), lastAttempt (date), createdAt, and updatedAt. Payment records purchases and associates a User with a Lesson: `_id`, `userId`, `lessonId`, amount (number), currency (string), status (string), transactionDate (date), transactionId (string, optional), unlockedDifficulty (string), createdAt, and updatedAt. Lesson is a standalone resource containing `_id`, title, description, content, difficulty, tags (array), quiz (embedded array of quiz items), createdAt, and updatedAt. Each embedded Quiz item includes question (string), options (array), and correctAnswer (number). Reward stores badge definitions and metadata: `_id`, name, pointsRequired (number), description, icon, createdAt, and updatedAt.

Work Breakdown Structure (WBS)

The Work Breakdown Structure organises the project into eight top-level phases, each decomposed into specific deliverable tasks. Current status is tracked across all tasks.

Phase	Task	Status
Planning & Requirements	Project scope and objectives definition	Completed
Planning & Requirements	Functional and non-functional requirements	Completed
Planning & Requirements	Artefact design and system diagrams	Completed
Design & Architecture	Overall system Architecture (MERN stack)	Completed
Design & Architecture	Database schema and ERD	Completed
Design & Architecture	API endpoint specification and design	Completed
Design & Structure	Security and authentication design	Completed
Backend Development	User authentication module (JWT, Google OAuth)	Completed
Backend Development	Core data models (User, Quest, Lesson, Progress)	Completed
Backend Development	Quest and Lesson CRUD API routes	Completed
Backend Development	Payment integration and webhook handling	Completed
Backend Development	Gamification engine (XP, badges, levels, streaks)	Completed
Backend Development	Email verification service (SMTP/Nodemailer)	Completed
Frontend Development	Landing page and authentication views	Completed
Frontend Development	User dashboard and statistics view	Completed
Frontend Development	Admin panel for quest and lesson management	Completed
Frontend Development	Quest browsing and search functionality	Completed

Phase	Task	Status
Frontend Development	Lesson player and quiz interface	Completed
Frontend Development	Payment checkout and content unlock flow	Completed
Content & Gamification	Microskill quest design (Git, JavaScript, SQL, Docker)	In Progress
Content & Gamification	Lesson and quiz content authoring	Completed
Content & Gamification	Badge visuals and reward structure	Completed
Testing & QA	Backend unit and integration testing	Completed
Testing & QA	API endpoint testing (Postman)	Completed
Testing & QA	UI and acceptance testing	In Progress
Testing & QA	Payment flow and security testing	Completed
Deployment & DevOps	Frontend build and deployment (Vercel)	Pending
Deployment & DevOps	MongoDB Atlas setup and data migration	Pending
Deployment & DevOps	Domain, SSL and webhook configuration	Pending
Documentation & Handover	Artefact design report	Completed
Documentation & Handover	Final project report	Completed
Documentation & Handover	API documentation and usage examples	In Progress

The WBS confirms that the project is in an advanced implementation phase. Core structure, backend services, authentication, payment integration, gamification engine, and all major frontend components are fully completed. Remaining work focuses on content authoring, future deployment, and documentation completion.

Wireframes

Wireframing was primarily focused on the authentication interfaces, specifically the login and registration screens. The remaining system interfaces are documented through structured UI descriptions and implementation notes to represent their intended layout, functionality, and user interaction behavior.

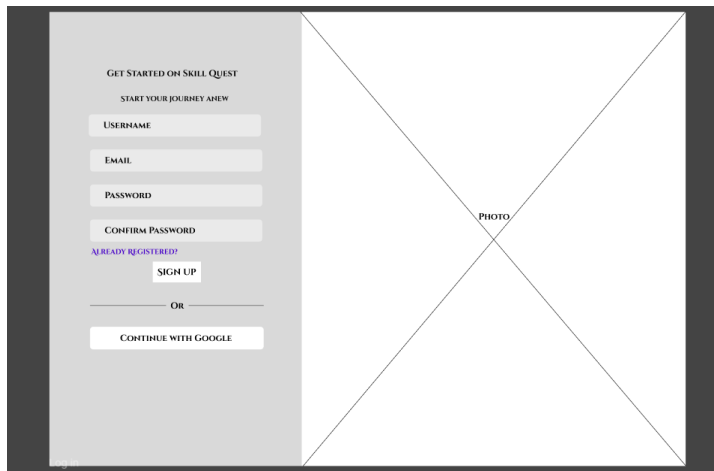


Figure 14: Sign up page wireframe

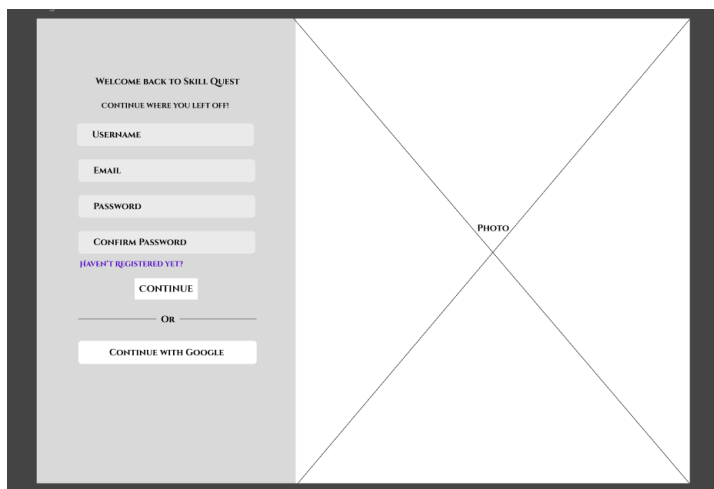


Figure 15: Login page wireframe

Software / Project Methodology

Explanation of Used Methodology

The project was developed following the Agile software development methodology, specifically using an iterative sprint-based approach inspired by Scrum. The development was organised, each targeting a defined set of user stories and deliverables. At the beginning of each sprint, tasks were selected from a product backlog based on priority and estimated effort. At the end of each sprint, a review was conducted to assess completed features, identify bugs, and reprioritise the backlog for the next cycle.

The overall project lifecycle followed four main agile phases: requirements and planning, system design, iterative development, and testing. Backend routes were generally implemented before frontend integration so that API functionality could be tested independently using Postman.

Version control was maintained using Git with a branching strategy: a main branch for stable project versions, a develop branch for active development, and feature branches for individual features (e.g., feature/payment-integration, feature/admin-dashboard).

Why Agile Was Used

Agile was selected over waterfall or plan-driven methodologies for several reasons specific to this project's context:

- **Evolving requirements:** The feature set evolved throughout development as the system design became more refined. Agile's iterative approach allowed new requirements (e.g., the addition of webhook-based payment confirmation) to be incorporated without requiring a complete redesign.
- **Solo development context:** As a sole developer, short sprint cycles allowed regular self-review of progress and course corrections without the overhead of formal team ceremonies.

- Early feedback through testing: By delivering working software at the end of each sprint, it was possible to conduct API testing and UI testing progressively rather than waiting until all features were complete, enabling early detection of integration issues.
- Risk management: High-risk components (payment integration, authentication, admin access control) were scheduled in early sprints to ensure sufficient time for debugging and redesign if needed.
- Flexibility in scope: The WBS distinguished between must-have features (authentication, quest/lesson CRUD, quiz flow, rewards) and nice-to-have features (full adaptive gamification, email notifications). Agile facilitated dynamic scope management when time constraints required prioritisation decisions.

Tools and Technology

Category	Technology / Tool	Purpose
Frontend Framework	React 19 + Vite	Component-based UI with fast HMR development server
Styling	Tailwind CSS + PostCSS	Utility-first responsive styling
Animations	Framer Motion	UI transition and gamification animations
Icons	Lucide React, React Icons	Consistent iconography throughout the UI
Routing	React Router DOM	Client-side page routing
HTTP Client	Axios	API request management with base URL configuration
Google Auth (Frontend)	@react-oauth/google	Google OAuth consent flow
Backend Runtime	Node.js + Express.js	REST API server and middleware handling
Database	MongoDB + Mongoose	Document-oriented data persistence and ODM

Category	Technology / Tool	Purpose
Authentication	Passport.js + JWT	Strategy-based auth with token issuance
Email	Nodemailer	Transactional email for account verification
Version Control	Git + GitHub	Source control and collaboration
API Testing	Postman	Manual API endpoint testing and request collection
Database Inspection	MongoDB Compass	Visual query interface for database verification
Development OS	Windows 11	Local development environment

Testing Approach

Explanation of Used Approach

The testing approach adopted for this project combined multiple testing types executed at different phases of development. The primary testing methods were:

- **Function Level Testing:** Individual backend controllers, service functions, and API routes were tested independently during development before full system integration.
- **Integration Testing:** Verification of interactions between the frontend, backend API, and MongoDB database across complete request-response cycles.
- **Functional Testing:** End-to-end validation of complete user workflows from authentication through quest completion and reward awarding.
- **Security Testing:** Targeted testing of authentication mechanisms, JWT validation, role-based access enforcement, and protection of admin-only endpoints.
- **User Acceptance Testing (UAT):** Manual scenario-based testing aligned with user stories to confirm that the system behaves as expected from the perspective of both learner and admin users.

The primary tooling was Postman for API endpoint testing (sending HTTP requests with custom headers, bodies, and authentication tokens) and MongoDB Compass for database state verification following test execution.

Why This Approach Was Used

A manual, API-first testing approach was selected for several practical reasons:

- **Early backend stability:** Testing API endpoints directly in Postman before connecting the frontend allowed rapid iteration on the backend without front-end complexity obscuring root causes of failures.
- **Authentication verification:** Role-based access control and JWT validation require precise control over request headers. Postman's header configuration and environment variable support made it the most efficient tool for verifying auth workflows.
- **Database state confirmation:** MongoDB Compass provided immediate visual confirmation that data was being correctly persisted, updated, and queried after API operations — critical for verifying payment record creation and progress tracking.
- **Scope-appropriate rigour:** Automated test frameworks (Jest, Supertest) would have added value in a team or CI/CD environment but introduced setup overhead disproportionate to the solo project scope. Manual testing provided sufficient rigour for the development context while maintaining velocity.

Test Execution Evidence

The following test cases were executed and their outcomes recorded. Screenshots of Postman requests and MongoDB Compass database states are included as evidence of test execution

TC-01: User Login

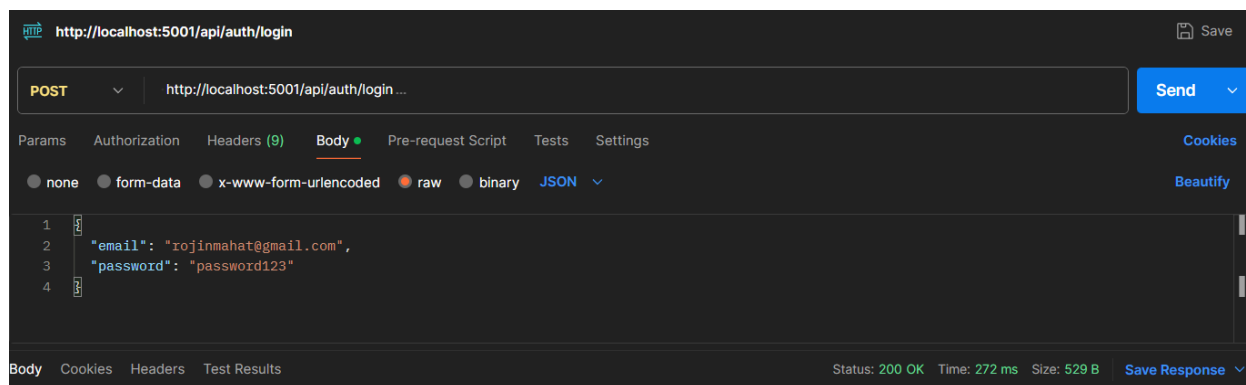


Figure 16: test case - user login

Field	Description
Objective	Verify registered users can log in and receive a JWT token
Precondition	User account exists in the database (registered previously)
Input	POST /api/auth/login with JSON body: { email: 'rojinmahat@gmail.com', password: 'password123' }
Expected Result	HTTP 200 OK; response body contains token (JWT string) and user object with id, name, role
Actual Result	HTTP 200 OK; JWT token and user object returned as expected
Status	PASS

TC-02: Access Beginner Lesson (Free)

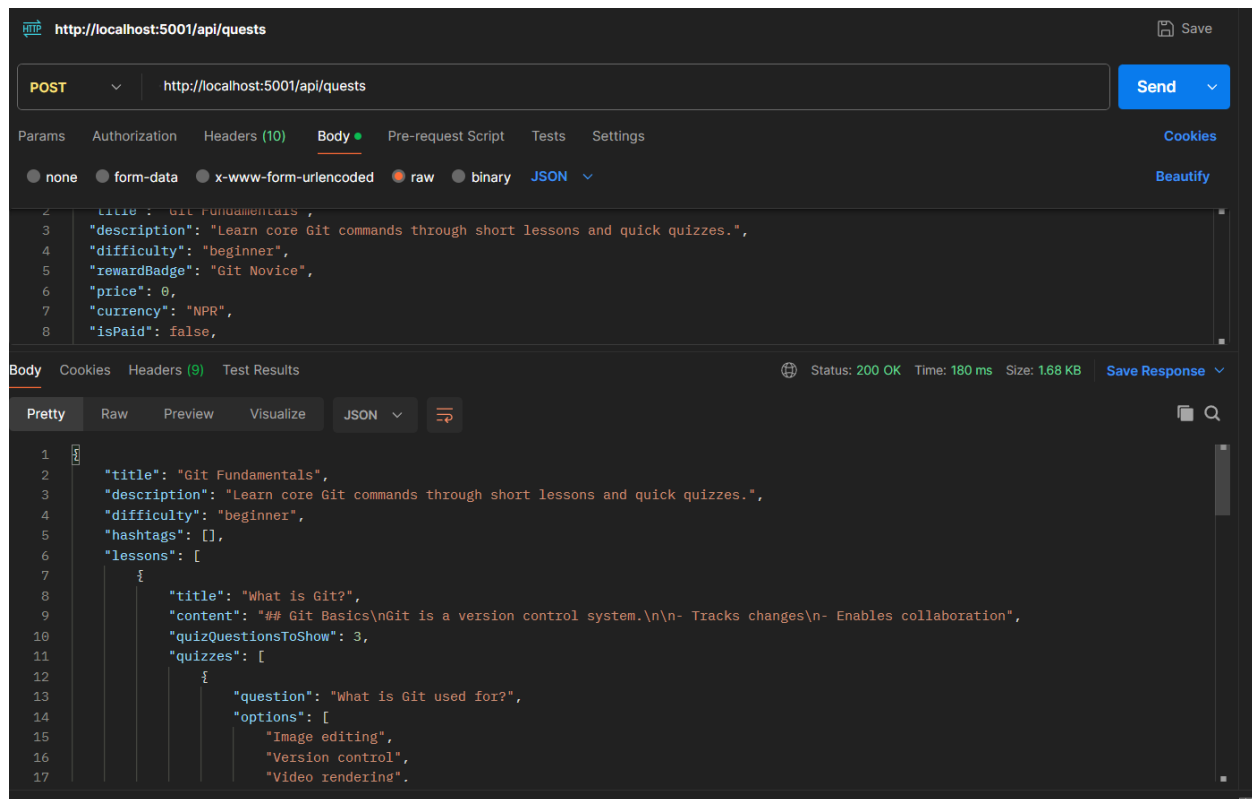


Figure 17: test case: accessing beginner free lessons using user/admin token

Field	Description
Objective	Verify free Beginner lessons are accessible without payment
Precondition	User authenticated; Beginner quest exists in database
Input	GET <code>/api/access?lessonId={id}&difficulty=beginner</code> with valid learner JWT
Expected Result	HTTP 200 OK; AccessGranted response; lesson content available
Actual Result	HTTP 200 OK; lesson loaded without payment flow
Status	PASS

TC-03: Access Restricted Lesson Without Payment

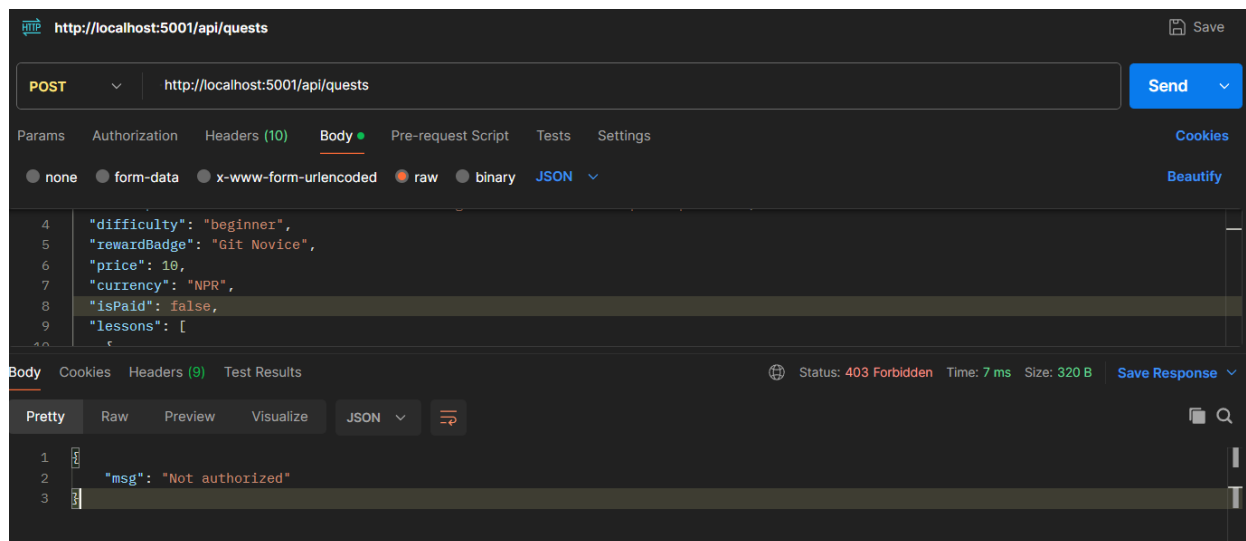


Figure 18: test case: accessing restricted content without payment

Field	Description
Objective	Prevent access to Intermediate/Advanced content without payment
Precondition	User authenticated; Intermediate quest exists; user has not paid
Input	GET /api/access?lessonId={id}&difficulty=intermediate with valid learner JWT
Expected Result	Access denied; payment prompt shown in UI
Actual Result	System correctly redirected to payment flow
Status	PASS

TC-04: Payment-Based Unlocking



Figure 19: ids of contents purchased by the user

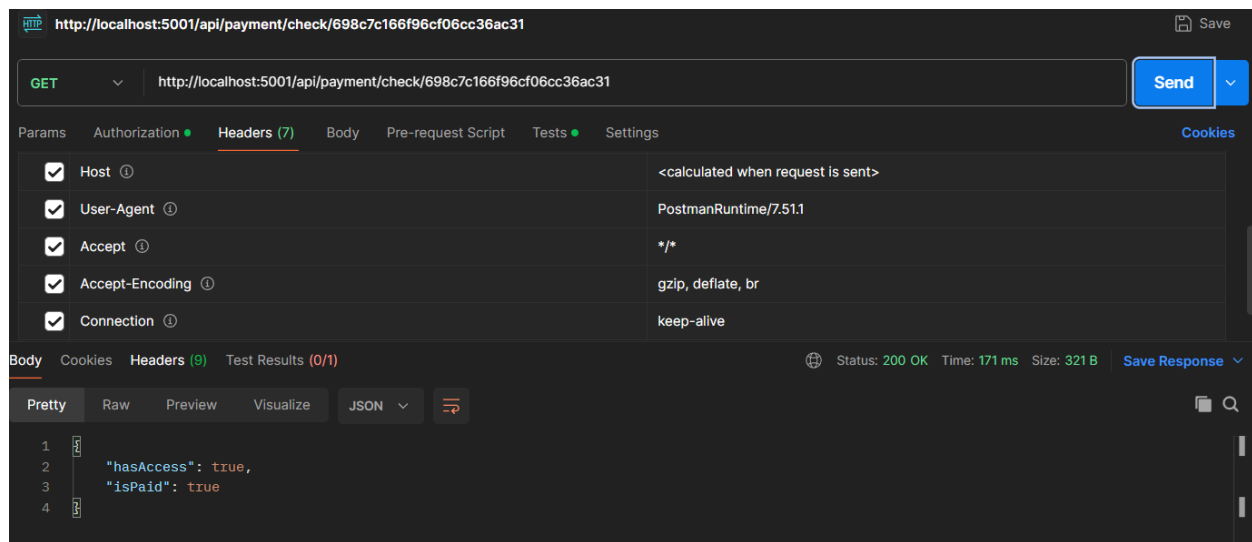


Figure 20: test case - the user has access to the paid quest

Field	Description
Objective	Verify paid content is unlocked after successful payment
Precondition	User authenticated; payment sandbox environment active
Input	Complete payment flow; webhook delivers success callback
Expected Result	Payment record created with status: completed; user gains access to locked content
Actual Result	Payment record persisted correctly; content unlocked on re-query

Field	Description
Status	PASS

TC-05: Lesson Completion — Correct Answer

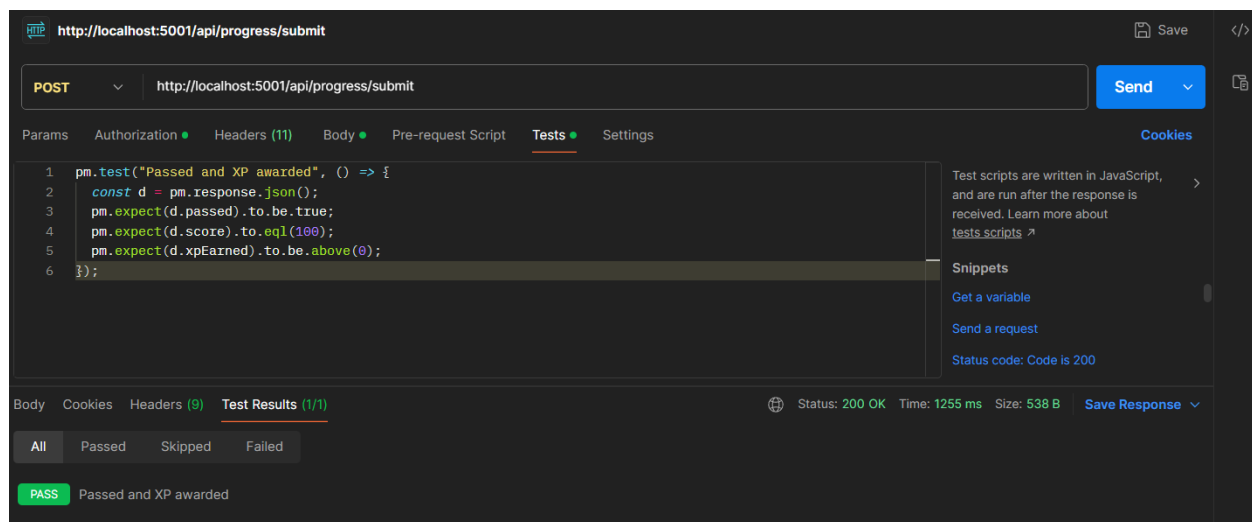


Figure 21: test case - lesson completion with correct answers

Field	Description
Objective	Validate that correct quiz answers trigger XP award and progress update
Precondition	Lesson unlocked; user on quiz screen; valid learner JWT
Input	POST /api/progress/submit with lessonId, questId, answers, and optionally questionIndexes
Expected Result	Progress record updated; XP incremented in user document; success feedback shown
Actual Result	Progress updated and rewards awarded correctly
Status	PASS

TC-06: Lesson Completion — Incorrect Answer

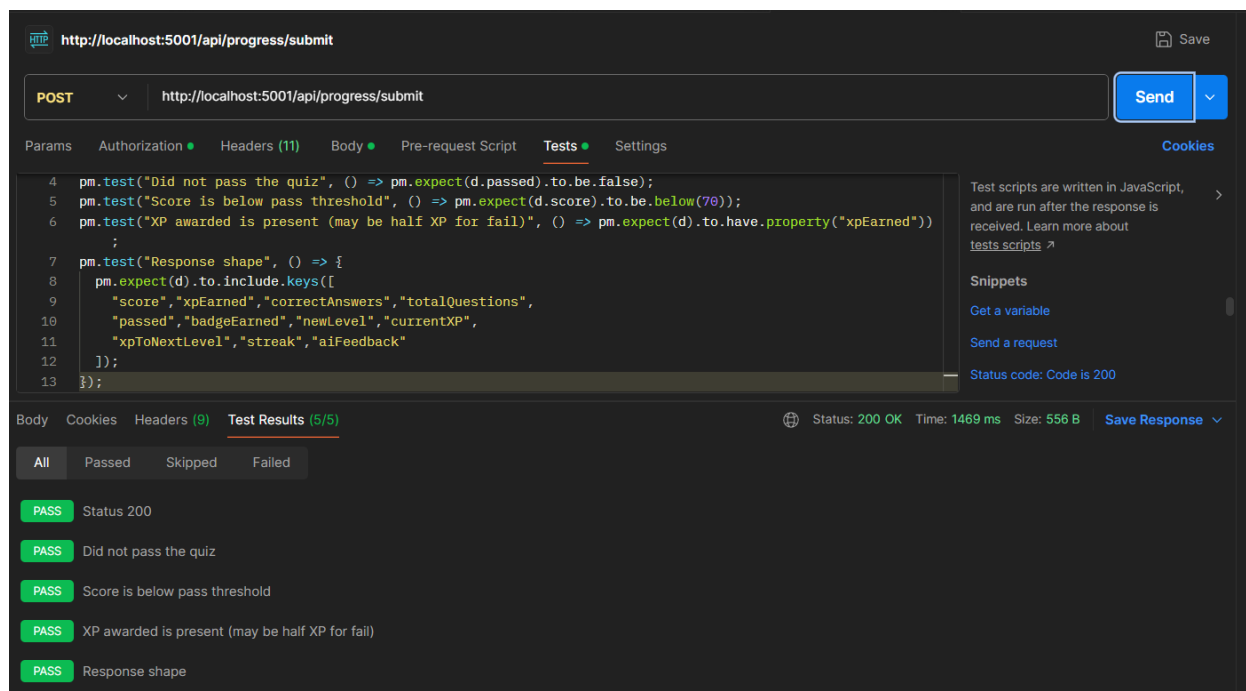


Figure 22: test case - lesson completion with incorrect answers

Field	Description
Objective	Validate retry logic on incorrect quiz answer
Precondition	Lesson unlocked; user on quiz screen
Input	Incorrect answer selected and submitted
Expected Result	Feedback shown with correct answer highlighted; Retry Lesson option presented
Actual Result	Immediate feedback and retry option displayed correctly
Status	PASS

TC-07: Admin Creates Quest/Lesson

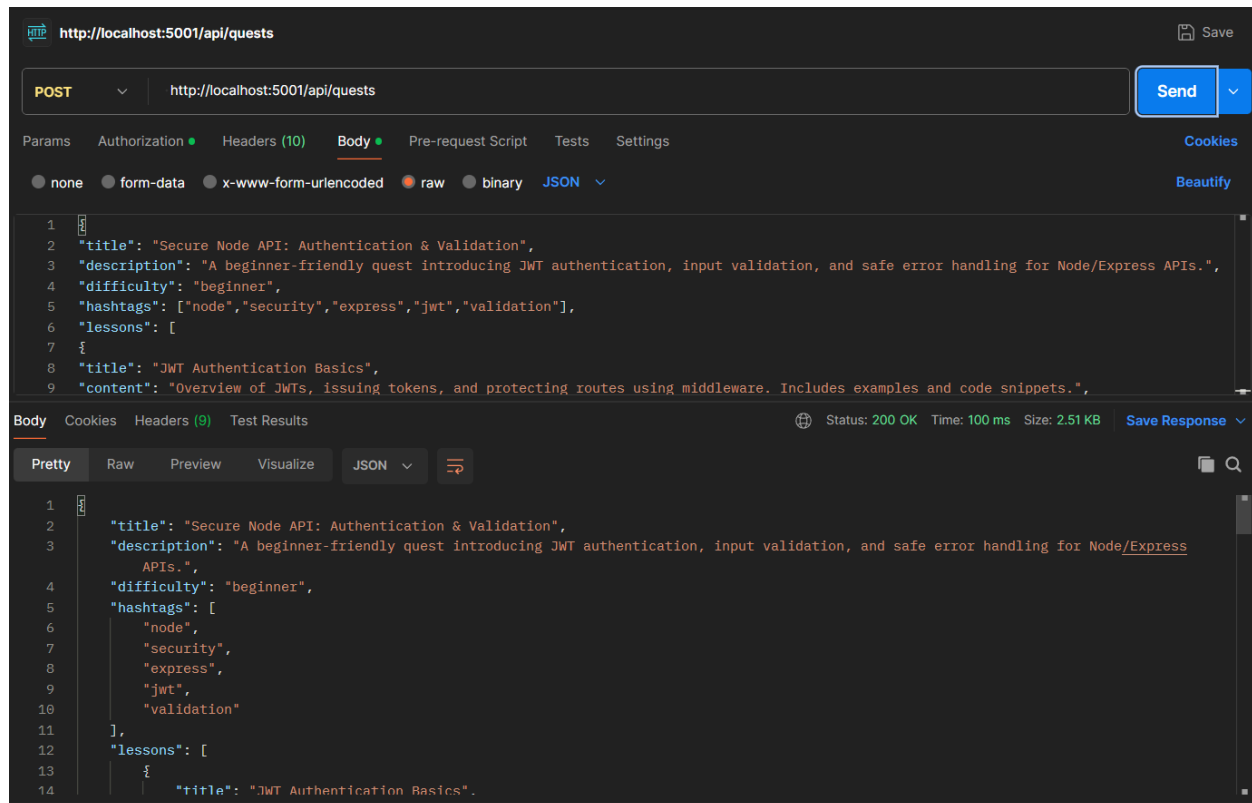


Figure 23: test case - adding a quest as admin



Figure 24: mongodb compass showing the recently added quest

Field	Description
Objective	Verify admin can create new quest with embedded lessons and quizzes
Precondition	Admin authenticated; admin JWT available
Input	POST /api/quests with admin Bearer token and quest JSON body including lessons array

Field	Description
Expected Result	HTTP 200 OK; quest document created in MongoDB quests collection
Actual Result	Quest successfully created and verified in MongoDB Compass
Status	PASS

TC-08: Role-Based Access Control

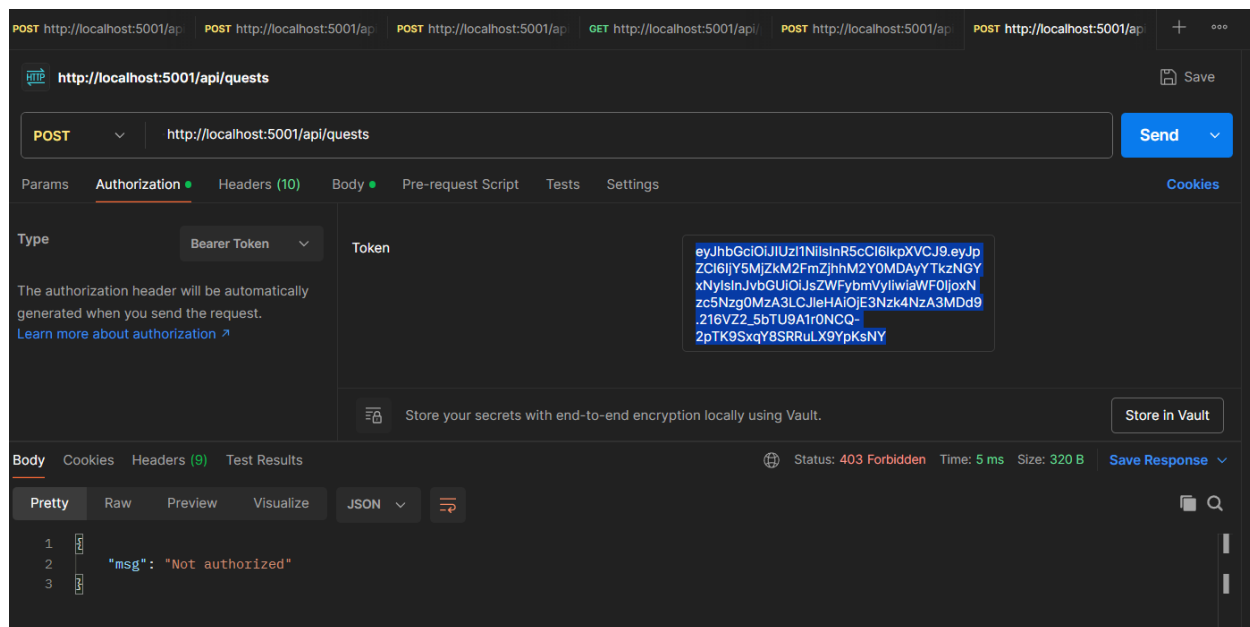
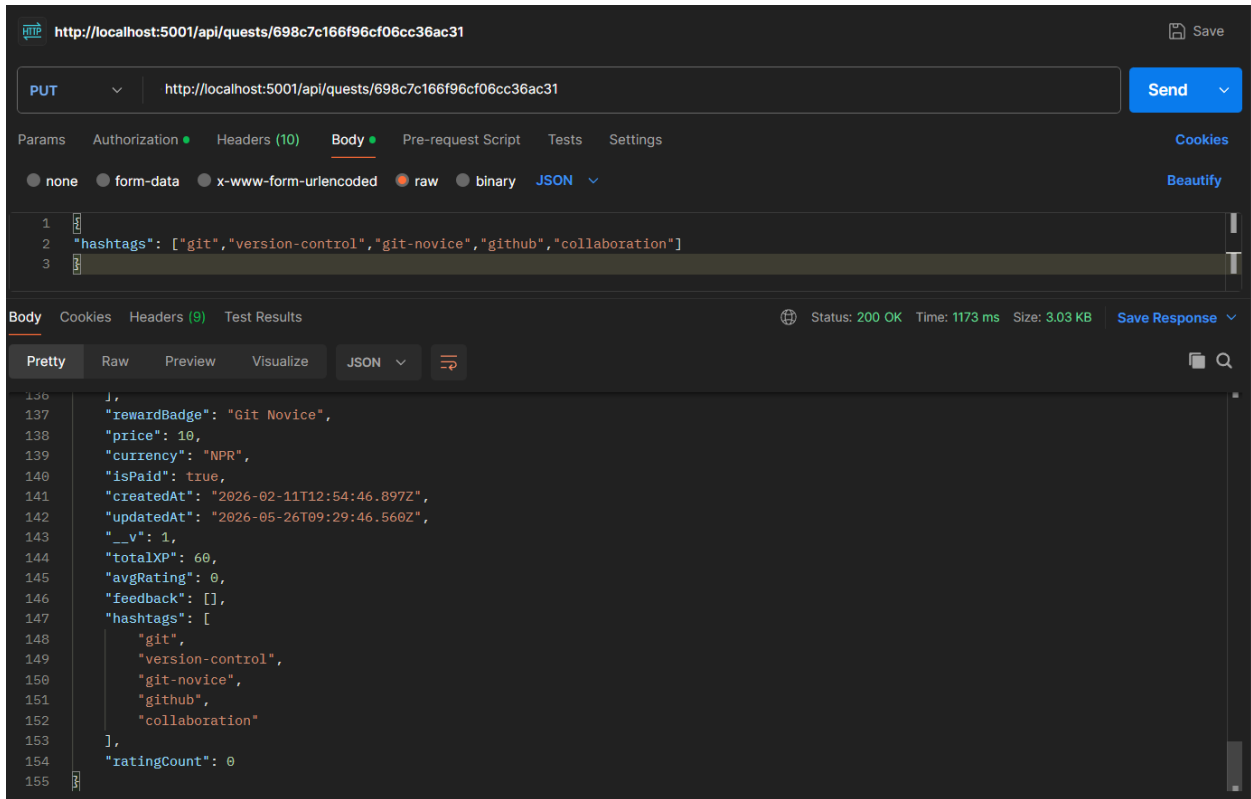


Figure 25: test case: role based access control

Field	Description
Objective	Verify learners cannot access admin-only endpoints
Precondition	Learner authenticated; non-admin JWT available
Input	POST /api/quests with learner Bearer token (non-admin role)
Expected Result	HTTP 403 Forbidden; msg: “Not authorized”
Actual Result	HTTP 403 Forbidden returned as expected
Status	PASS

Test case – update quest as an admin



The screenshot shows a REST client interface with a PUT request to `http://localhost:5001/api/quests/698c7c166f96cf06cc36ac31`. The request body is a JSON object with a `hashtags` array. The response is a JSON object with various fields including `rewardBadge`, `price`, `currency`, `isPaid`, `createdAt`, `updatedAt`, `__v`, `totalXP`, `avgRating`, `feedback`, `hashtags`, and `ratingCount`.

Request URL: `http://localhost:5001/api/quests/698c7c166f96cf06cc36ac31`

Method: PUT

Body (JSON):

```
1 {
2   "hashtags": ["git", "version-control", "git-novice", "github", "collaboration"]
3 }
```

Response (JSON):

```
136 },
137 "rewardBadge": "Git Novice",
138 "price": 10,
139 "currency": "NPR",
140 "isPaid": true,
141 "createdAt": "2026-02-11T12:54:46.897Z",
142 "updatedAt": "2026-05-26T09:29:46.560Z",
143 "__v": 1,
144 "totalXP": 60,
145 "avgRating": 0,
146 "feedback": [],
147 "hashtags": [
148   "git",
149   "version-control",
150   "git-novice",
151   "github",
152   "collaboration"
153 ],
154 "ratingCount": 0
155 }
```

Figure 26: test case - updating existing quest as an admin


{title: "Git Fundamentals"}

+ ADD DATA
UPDATE
DELETE
EXPORT DATA
EXPORT CODE


```

_id: ObjectId('698c7c166f96cf06cc36ac31')
title: "Git Fundamentals"
description: "Learn core Git commands through short lessons and quick quizzes."
difficulty: "beginner"
▶ lessons: Array (3)
rewardBadge: "Git Novice"
price: 10
currency: "NPR"
isPaid: true
createdAt: 2026-02-11T12:54:46.897+00:00
updatedAt: 2026-05-26T09:29:46.560+00:00
__v: 1
totalXP: 60
avgRating: 0
▶ feedback: Array (empty)
▼ hashtags: Array (5)
  0: "git"
  1: "version-control"
  2: "git-novice"
  3: "github"
  4: "collaboration"
ratingCount: 0

```

Figure 27: mongodb showing the updated quest hashtags section

Field	Description
Test Case ID	TC-Update-Quest-Hashtags-01
Objective	Verify that an admin can update hashtags for an existing quest to improve discoverability
Precondition	Admin user is authenticated and verified; valid admin JWT is available in {{ADMIN_BEARER}}; quest with ID 698c7c166f96cf06cc36ac31 exists
Input	PUT /api/quests/698c7c166f96cf06cc36ac31 with JSON body: { "hashtags": ["git","version-control","git-novice","github","collaboration"] } using admin Bearer token
Expected Result	HTTP 200 OK; response body contains updated quest object with hashtags exactly matching the provided array
Actual Result	To be filled after execution (expected: hashtags updated successfully)
Postcondition / Verification	GET request to the same quest ID returns the updated hashtags persistently stored in the database
Status	PASS / FAIL (to be determined after test execution)

Conclusion

This project has successfully delivered a functional gamified micro-learning platform for technical micro-skills, built on the MERN stack with a full suite of learner and administrative features. The platform addresses the central problem of low engagement and poor retention in self-directed online technical learning by combining evidence-based gamification mechanics, XP, badges, streaks, and progress tracking, with a structured microlearning curriculum, mandatory quiz-based assessment, and immediate feedback loops.

All primary objectives were met. A role-based system was implemented supporting both learner and administrator workflows. Structured learning content was organised into quests and lessons with difficulty tiers. Quiz assessment for active retrieval was integrated throughout the lesson flow. Progress tracking and reward allocation were implemented and verified. Administrative tools for content management and user monitoring were delivered. Payment-based unlocking of premium content was implemented with webhook verification. The system was built to be modular, maintainable, and aligned with JavaScript/JSX, Express, and MongoDB conventions.

Testing confirmed that the core functional requirements like authentication, role-based access control, lesson access, quiz evaluation, reward awarding, and payment-based unlocking all performed as specified. API tests executed in Postman with MongoDB Compass verification provided documented evidence of correct system behaviour.

The literature review confirmed that the design approach is well grounded in educational research. Gamification mechanics such as XP rewards, badges, progress tracking, and quiz based progression were aligned with learner performance and mastery.

The project contributes a practical reference implementation of gamified micro-learning that can be extended with adaptive personalisation, spaced repetition scheduling, and a full technical content curriculum in future development phases.

Critical Evaluation

This chapter reflects critically on the overall project, evaluating what was achieved, what limitations remain, what challenges were encountered, and what improvements could be made in future iterations.

Achievement Against Objectives

The project delivered all high-priority objectives. Authentication (email/password + Google OAuth), quest/lesson CRUD, quiz assessment, reward awarding, progress tracking, role-based access control, and payment-based unlocking were all implemented and tested. The system architecture is well-structured, with clear separation between frontend components, backend routes, middleware, and data models.

Medium-priority objectives — specifically the adaptive gamification engine and email verification service — were implemented in their foundational forms but not fully validated in production-like conditions. Full adaptive personalisation based on learner profile data remains on the development roadmap.

Challenges Encountered

Payment Integration Complexity

Integrating the third-party payment provider with webhook-based confirmation was the most technically challenging component. Webhook delivery in local development environments requires tunnelling tools (e.g., ngrok) to expose the local server to the internet, introducing an additional dependency. The payment workflow was tested in a sandbox development environment rather than a live public payment environment. Payment state management across asynchronous webhook callbacks required careful handling of race conditions between the frontend polling for access confirmation and the backend webhook handler updating the payment record.

JWT Token Management Across OAuth Flows

Reconciling JWT issuance across two authentication paths — email/password login and Google OAuth callback — required careful coordination between Passport strategies and the token issuance service. Early in development, token claims were inconsistent between the two flows, causing intermittent role-based access failures that were challenging to debug without dedicated logging.

Embedded Document vs. Reference Trade-offs

The decision to embed lessons and quizzes directly within quest documents (rather than referencing them by ID) optimised read performance for quest browsing but complicated update operations. Editing a single lesson within a quest requires targeting the correct array element by index, which proved error-prone in the admin dashboard UI and required careful validation in the backend update handlers.

Scope Management

As development progressed, the scope of desirable features expanded — adaptive quiz difficulty, learner leaderboards, social features, and a full technical content curriculum were all identified as high-value additions. Maintaining focus on the MVP feature set required active prioritisation decisions and acceptance that some planned features would remain on the roadmap rather than in the delivered system.

Comparison to Similar Platforms

Compared to Codecademy, the platform's admin content management capabilities are a significant differentiator — content is fully controlled by the platform operator rather than being locked to a fixed curriculum. However, Codecademy's interactive in-browser code execution environment, which provides real-time syntax feedback, is substantially more sophisticated than the multiple-choice quiz model implemented here.

Compared to Duolingo, the gamification depth is less mature — Duolingo's adaptive spaced repetition scheduling, lives system, and social leaderboards represent years of product iteration

that cannot be replicated within a single-developer academic project. However, the platform's focus on technical micro-skills and admin-managed content fills a niche that Duolingo does not address.

Compared to Khan Academy, the platform provides a more engaging motivational structure through XP, badges, and streaks, but lacks the breadth of content and the mastery-retesting system that makes Khan Academy pedagogically rigorous at scale.

Future Improvements

- Adaptive gamification: Implement learner profiling to tailor visible gamification elements (hide leaderboards for non-competitive users; prioritise mastery tracking for performance-avoidance learners).
- Spaced repetition scheduling: Add a lesson revisit scheduler that surfaces lessons for review based on elapsed time since last correct answer, implementing a basic SM-2 or Leitner-style spacing algorithm.
- Interactive code execution: Replace multiple-choice quizzes with sandboxed code execution challenges for programming micro-skills, providing richer and more authentic assessment.
- Full content curriculum: Author a complete set of technical micro-skill quests for Git, JavaScript, SQL, and Docker, populating the platform with fully functional learning content.
- Future deployment: Future work may include deploying the platform using services such as Vercel and MongoDB Atlas to make the system publicly accessible.
- Automated testing: Introduce a Jest + Supertest test suite for backend repeated testing and Cypress for end-to-end UI testing in automated testing workflow integration.

Evidence of Project Management

Log Sheet

The following supervisor meeting log summarises key interactions and milestones throughout the project lifecycle.

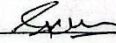
Faculty of Science and Engineering School of Mathematics and Computer Science		 UNIVERSITY OF WOLVERHAMPTON
PROJECT MANAGEMENT LOG		
First Name: <u>Rojin</u>	Surname: <u>Mahol</u>	
Student Number: <u>2412733</u>	Supervisor: <u>Saddam Khan</u>	
Project Title: <u>SkillQuest (Gamified Learning app)</u> Month: <u>November</u>		
What have you done since the last meeting		
The backend was successfully developed using MERN stack, featuring JWT-based authentication and MongoDB integration. User registration, login, and user-data retrieval end-point routes were implemented and tested. On the frontend, responsive login and registration interfaces were developed using React and tailwindcss, maintaining a modern and consistent design theme. Token handling and auth.-based redirection were integrated. Initial development of user dashboard was initiated. Also created MongoDB models, and a landing page		
What do you aim to complete before the next meeting		
Complete dashboard interface, connect frontend to backend APIs for data display, implement quest browsing page, create sample quest content.		
Supervisor comments		
> Frontend & backend part for user dashboard > And also work on artifacts design		
We confirm that the information given in this form is true, complete and accurate.		
Student Signature: <u></u>	Date: <u>2025/11/16</u>	
Supervisor Signature: <u></u>	Date: <u>23/11/2025</u>	

Figure 28: Log sheet 1

Note: this is only one of the log sheets maintained throughout the project. The entirety of the log sheet is in the appendix section of the report

Gantt Chart

The Gantt chart below summarises the project timeline across its major phases. All dates are approximate sprint-level estimates.

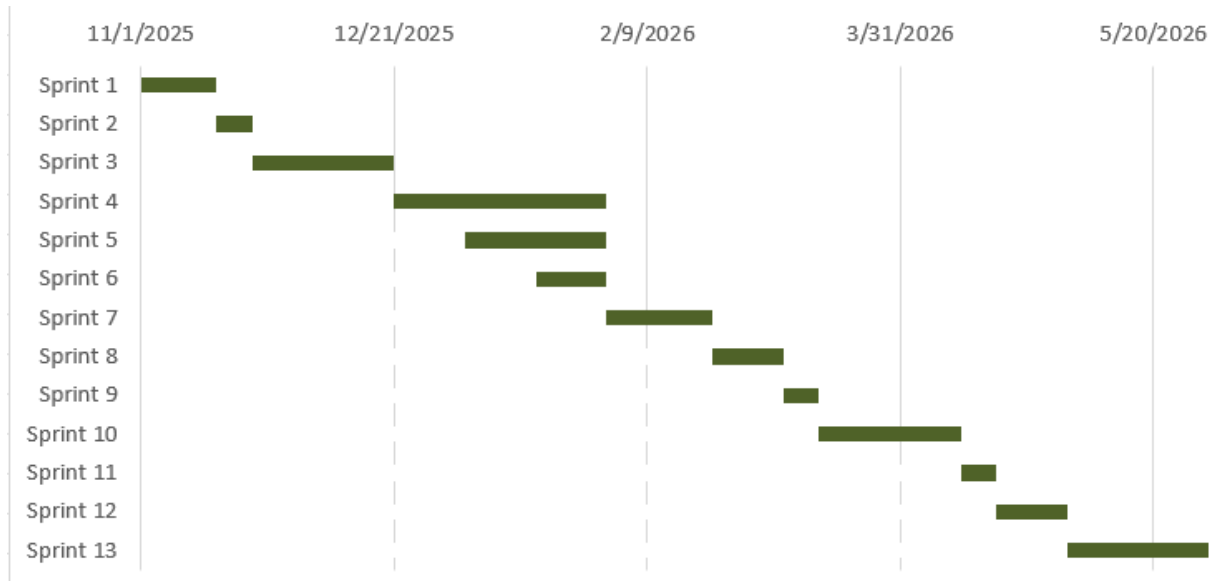


Figure 29: Gantt-chart of the project

Risk Management

The sheet below shows the risks identified and managed across the project timeline.

Risk id	Risk Description	Category	Likelihood	Impact	Priority	Mitigation Strategy	Contingency Plan
R01	Scope creep - adding too many features	Project Management	High	High	Critical	Define clear MVP scope, use Agile sprints to manage changes, strict feature prioritization	Cut non-essential features, extend timeline if approved by supervisor
R02	Timeline delays due to technical complexity	Technical	Medium	High	High	Build buffer time into sprints, prioritize core features first, weekly progress reviews	Reduce feature scope, focus on MVP, request deadline extension
R03	Backend API failures or bugs	Technical	Low	Critical	High	Unit testing with Jest, API testing with Postman, code reviews	Implement robust error handling, maintain backup database snapshots
R04	Database connection issues or data loss	Technical	Low	Critical	High	Use MongoDB Atlas for reliability, implement regular backups, test connection handling	Switch to backup database instance, restore from recent backup
R05	Authentication security vulnerabilities	Security	Medium	Critical	Critical	Use bcrypt for password hashing, implement JWT properly, security testing	Immediate patching, review auth flow, consult security resources
R06	Frontend-backend integration problems	Technical	Medium	Medium	Medium	Regular integration testing, use consistent API contracts, Postman collection for testing	Debug systematically, check CORS settings, verify endpoint connections
R07	Poor user interface/experience	Design	Medium	Medium	Medium	User feedback sessions, usability testing with 5-10 users, iterative design improvements	Redesign problem areas based on feedback, consult UI/UX resources
R08	Lack of engaging gamification elements	Design	Medium	High	High	Research best practices, test with users, iterate based on feedback	Simplify gamification, focus on core mechanics (points, badges)
R09	Payment gateway integration failures	Technical	Medium	High	High	Use Stripe test mode extensively, follow documentation carefully, handle edge cases	Use simpler payment stub, mark as future enhancement
R10	Insufficient content (micro-skill lessons)	Content	High	High	Critical	Write 2-3 lessons per week starting early, reuse lesson templates	Reduce initial lesson count to 10-15, add more post-launch
R11	Figma AI generates unusable code	Technical	Low	Medium	Low	Review and test all generated code, modify as needed, use as starting point only	Code components manually, use Figma only for design inspiration
R12	Local Environment and Deployment Issues	Infrastructure	Medium	Medium	Medium	the application will be tested regularly in the local environment	the system can be restored using backup project files or using a stable local build
R13	Browser compatibility issues	Technical	Low	Low	Low	Test on major browsers (Chrome, Firefox, Safari, Edge), use standard web technologies	Document supported browsers, provide fallback UI
R14	Literature review quality insufficient	Academic	Low	Medium	Medium	Use recent peer-reviewed sources (2020+), properly cite all sources, get supervisor feedback	Add more sources, expand analysis, revise based on feedback
R15	Supervisor unavailability or miscommunication	Project Management	Medium	Medium	Medium	Schedule meetings in advance, send progress updates via email, document decisions	Proceed with best judgment, document reasoning, catch up at next meeting
R16	Burnout or time management issues	Personal	Medium	High	High	Realistic sprint planning, take regular breaks, maintain work-life balance	Reduce daily work hours, extend sprint length, seek support
R17	Health issues causing delays	External	Low	High	Medium	Maintain good health practices, have work-from-home setup ready	Notify supervisor immediately, adjust timeline, work remotely
R18	Technology stack changes mid-project	Technical	Low	High	Medium	Commit to chosen stack early, only change if critical issue arises	Assess impact, minimize changes to affected modules only
R19	Data validation and input sanitization gaps	Security	Medium	Medium	Medium	Implement input validation on both frontend and backend, test with invalid data	Add validation middleware, sanitize all user inputs
R20	Mobile responsiveness issues	Design	Medium	Low	Low	Tailwind responsive classes, test on multiple screen sizes regularly	Focus on desktop first, mark mobile as future work

Figure 30: Risk register

Risk Register Narrative

R01 – Scope Creep

There is a risk that additional features may continue to be added beyond the minimum viable product, increasing workload and affecting deadlines. This risk falls under project management and is considered both highly likely and highly impactful, making it a critical priority. To reduce this risk, the project will follow a clearly defined MVP approach, use sprint based prioritization, and limit unnecessary scope changes during development. If scope creep still becomes an issue, non-essential features will be removed and timeline extensions will only be considered with supervisor approval.

R02 – Timeline Delays

Certain technical tasks may take longer than expected, causing delays to the project schedule. This is categorized as a technical risk with medium likelihood and high impact. Buffer time will be included within the project plan, core features will be prioritized, and weekly progress reviews will be carried out to monitor development. If delays become unavoidable, the project scope may be reduced to focus only on the MVP or an authorized extension may be requested.

R03 – Backend Bugs and API Failures

Backend services or APIs may fail, crash, or return incorrect responses, potentially affecting system functionality. This technical risk has a medium likelihood and high impact. To minimize the risk, unit testing and integration testing will be conducted regularly, code reviews will be performed, and API endpoints will be validated using Postman throughout development. In case of failure, recent backups will be restored, bug fixes will be applied, and the application can be reverted to a previous stable version.

R04 – Database Connection Failure or Data Loss

Database errors or accidental data loss could affect application functionality and user information. Although the likelihood is low, the impact is considered critical. To minimise this risk, regular local database backups will be maintained, multiple copies of important project files will be stored, and database connections will be tested throughout development. If data loss occurs, the database can be restored from the most recent backup copy.

R05 – Authentication Security Issues

Weaknesses in the authentication system could expose user accounts or sensitive data. This security related risk has a medium likelihood but a critical impact. Secure password hashing using bcrypt, proper JWT implementation, and regular security testing will help reduce vulnerabilities. If a security issue is discovered, immediate patches will be applied, authentication flows will be reviewed, and additional security guidance will be consulted if required.

R06 – Frontend and Backend Integration Issues

Problems may arise when integrating frontend and backend systems, particularly if API contracts do not match correctly. This technical risk has medium likelihood and medium impact. To address this, API contracts will be clearly defined, integration testing will be performed using Postman collections, and regular end to end testing will be conducted. If integration issues occur, systematic debugging will be carried out, and endpoint or CORS configurations will be verified.

R07 – Poor UI and UX Design

An unclear or unattractive interface may negatively affect user engagement and overall usability. This design-related risk is considered medium in both likelihood and impact. The project will include usability testing with a small group of users to gather feedback and improve the interface iteratively. If the interface is still ineffective, problematic sections will be redesigned and UI or UX best practices will be reviewed.

R08 – Weak Gamification Features

The gamification system may fail to motivate or engage users effectively. This design risk has medium likelihood and high impact. To mitigate the issue, existing gamification techniques will be researched, tested with users, and refined based on feedback. If the features still prove ineffective, the platform will focus on simpler mechanics such as points, badges, and progress tracking to maintain clarity and engagement.

R09 – Payment Gateway Failure

Payment gateway integration may behave inconsistently or fail during transactions. This technical risk has medium likelihood and high impact. To reduce the risk, payment provider test

environments will be used extensively and integration documentation will be followed carefully. If integration problems persist, a simplified payment simulation may be used temporarily while the full payment feature is postponed as a future enhancement.

R10 – Insufficient Learning Content

The platform may not contain enough learning materials to provide meaningful value to users. This content related risk has both high likelihood and high impact, making it critical. To manage this risk, lessons will be created regularly using predefined templates and a structured content schedule will be followed. If content production falls behind, the system will launch with a smaller core set of lessons and additional content will be added gradually after launch.

R11 – Poor Figma AI Code Generation

AI generated code from Figma may not be reliable or suitable for production use. This technical risk has low likelihood and medium impact. Generated code will therefore only be treated as a starting reference and all outputs will be reviewed and tested manually. If the generated code proves unusable, components will be implemented manually while Figma continues to serve as a design reference tool.

R12 – Local Environment Configuration Issues

Problems with the local development environment or application configuration could prevent the system from running correctly during testing or demonstrations. This infrastructure related risk has medium likelihood and medium impact. To reduce this risk, the application will be tested regularly in the local environment and configuration files will be documented carefully. If issues occur, the system can be restored using backup project files or demonstrated using a stable local version of the application.

R13 – Browser Compatibility Issues

Certain browsers may not render features or layouts correctly, affecting user experience. This technical risk is considered low in both likelihood and impact. Testing will be carried out across major browsers including Chrome, Firefox, Safari, and Edge while following standard web

development practices. If compatibility issues remain, supported browsers will be documented and fallback interfaces will be provided where possible.

R14 – Poor Literature Review Quality

The literature review may fail to meet academic standards if sources are outdated or insufficiently analyzed. This academic risk has low likelihood and medium impact. To avoid this, recent peer reviewed sources published from 2020 onwards will be prioritized, proper referencing techniques will be used, and supervisor feedback will be sought regularly. If improvements are still needed, additional literature will be reviewed and the analysis will be revised accordingly.

R15 – Supervisor Unavailability

Communication delays or limited supervisor availability could affect project guidance and decision making. This project management risk has medium likelihood and medium impact. Meetings will therefore be scheduled in advance, regular updates will be shared, and key decisions will be documented. If communication gaps occur, development will continue based on documented reasoning until further feedback is available.

R16 – Burnout and Time Management Issues

Poor time management or personal burnout may slow project progress and reduce productivity. This personal risk has medium likelihood and high impact. To minimize this risk, realistic sprint planning will be followed, regular breaks will be taken, and work life balance will be maintained. If burnout becomes severe, daily workload may be reduced, sprint durations adjusted, and additional support sought if necessary.

R17 – Health Related Delays

Unexpected illness or health related issues may interrupt development activities. This external risk has low likelihood but high impact. To prepare for this possibility, remote working arrangements will remain available and healthy working practices will be maintained. If illness occurs, the supervisor will be informed promptly and the project timeline may be adjusted where required.

R18 – Mid Project Technology Stack Changes

Changing technologies during development may introduce delays and compatibility issues. This technical risk has low likelihood but high impact. The technology stack will therefore be finalized early in the project and unnecessary changes will be avoided. If a change becomes unavoidable, modifications will be limited to the affected modules and downstream effects will be evaluated carefully.

R19 – Weak Input Validation and Sanitization

Insufficient validation or sanitization may result in security vulnerabilities or poor data integrity. This security risk has medium likelihood and medium impact. To reduce the risk, validation will be implemented on both frontend and backend systems and tested using invalid or edge case inputs. If weaknesses are identified later, additional validation middleware will be added and stored data will be sanitised where necessary.

R20 – Mobile Responsiveness Issues

The application may not display properly on smaller screen sizes or mobile devices. This design related risk has medium likelihood and low impact. Responsive Tailwind utilities will be used during development and testing will be performed across multiple screen sizes. If responsiveness issues remain unresolved, the initial release may prioritise desktop usability while mobile improvements are scheduled for future updates.

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Appendix

User journey summary

Landing page

This is where every user starts



Figure 31: Site's landing page

Tutorial mode

Pressing the start your quest button takes the user to the Tutorial mode where they can learn the mechanics of the lesson contents and quizzes.

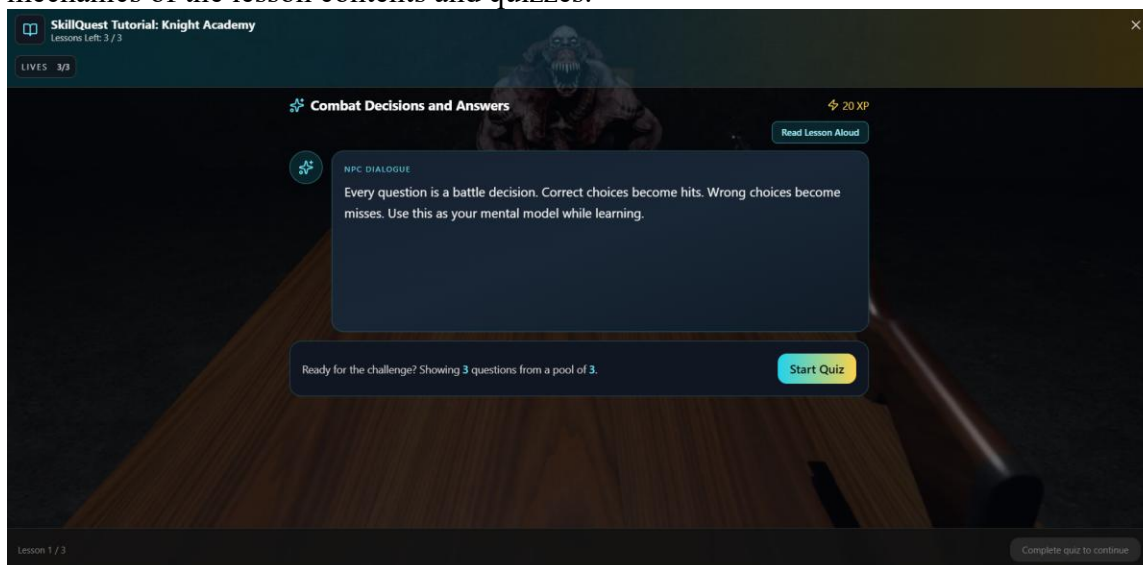


Figure 32: tutorial page

Login/Signup page

Completing and the tutorial or just choosing the login or sign up on the landing pages takes the user to the login or sign up pages respectively. The user may switch to either login or signup via the corresponding pages too.

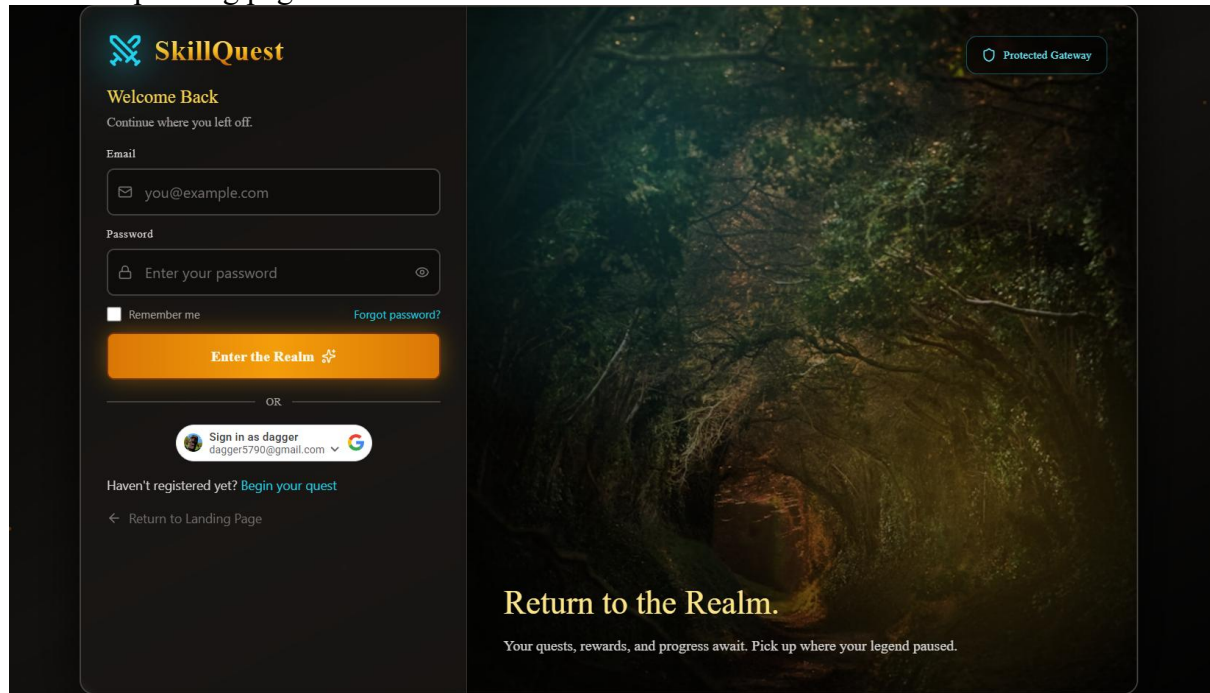


Figure 33: Login page

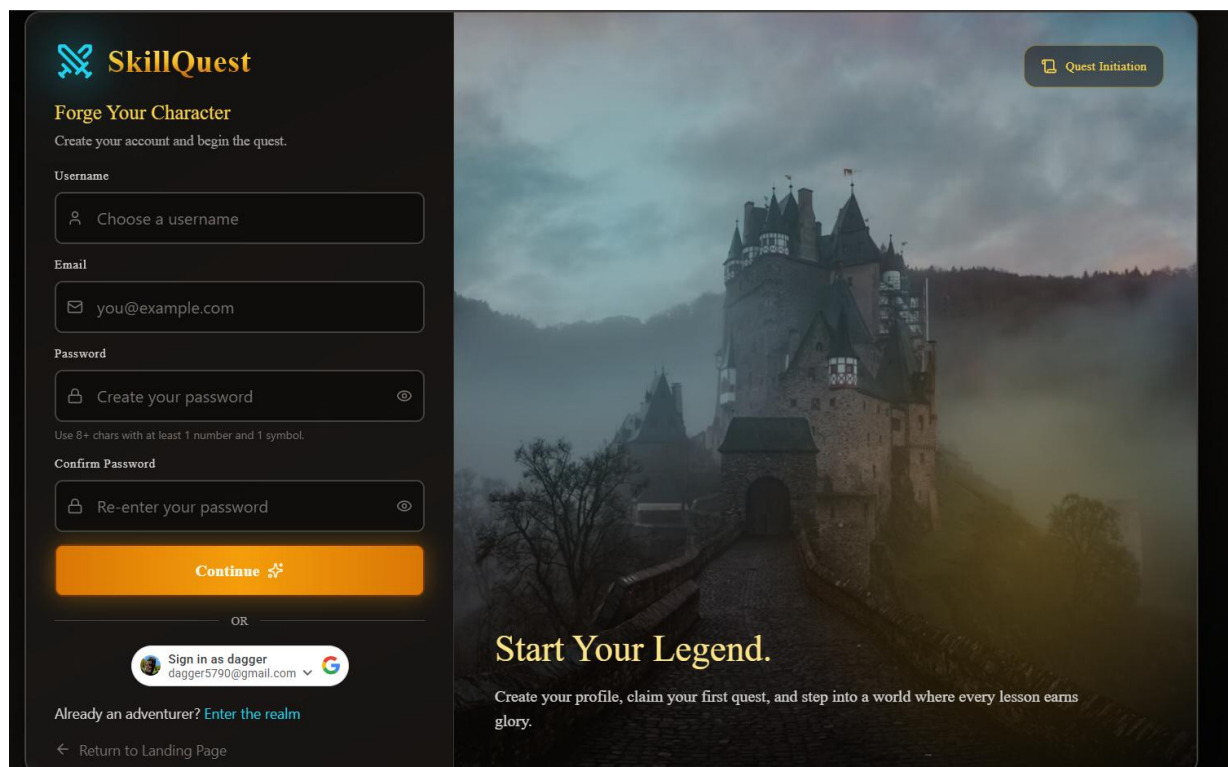


Figure 34: sign up page

User dashboard

A successful login takes the user to the learner's Dashboard.

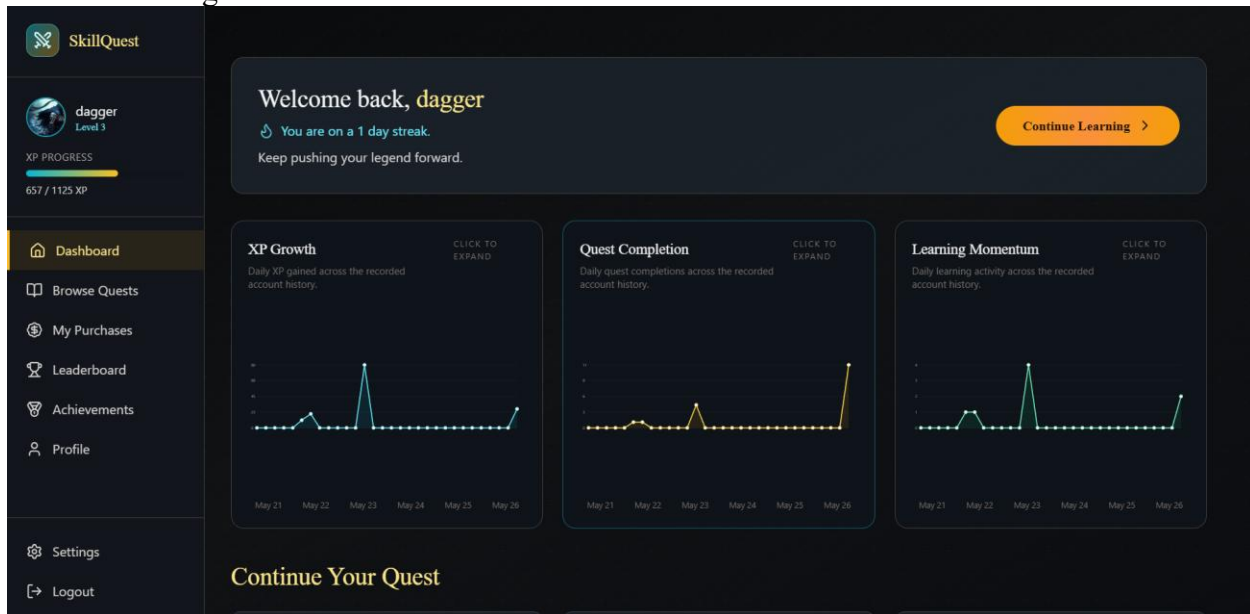
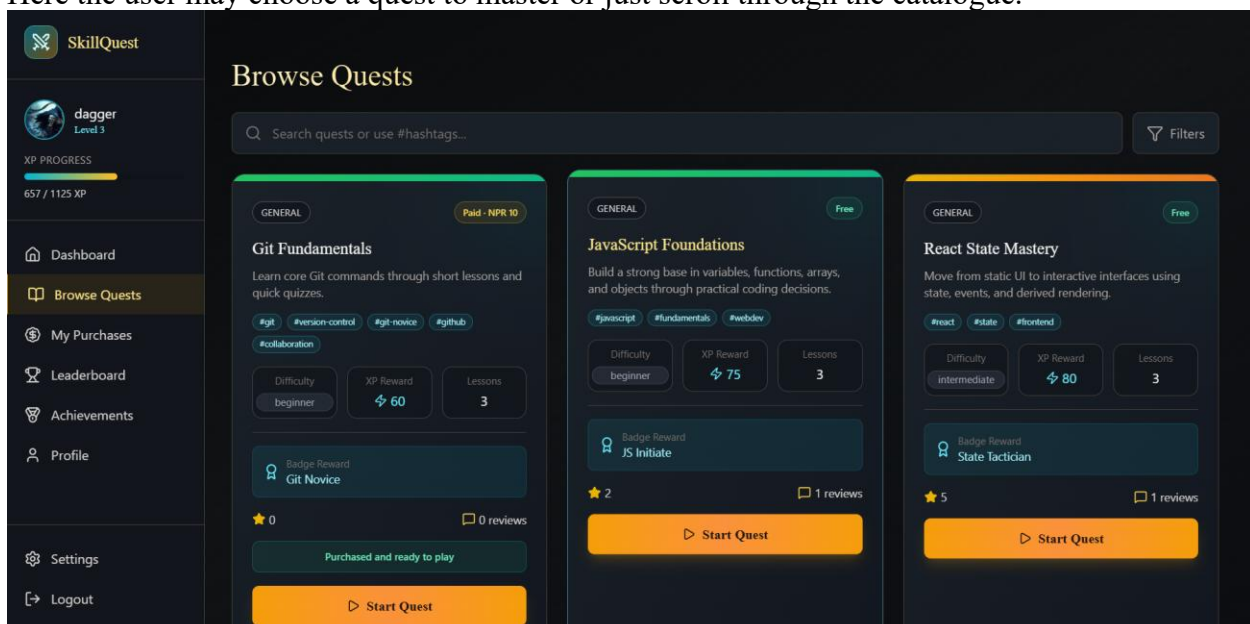


Figure 35: skillquest learner dashboard

Here, the user may choose to browse quests, check their purchases, check the leaderboards, achievements or set up their profile.

Quest catalogue

Here the user may choose a quest to master or just scroll through the catalogue.



Quest detail

After the user chooses a free quest or is done making a payment for a paid quest they may start the quest to go through its lessons and quizzes.

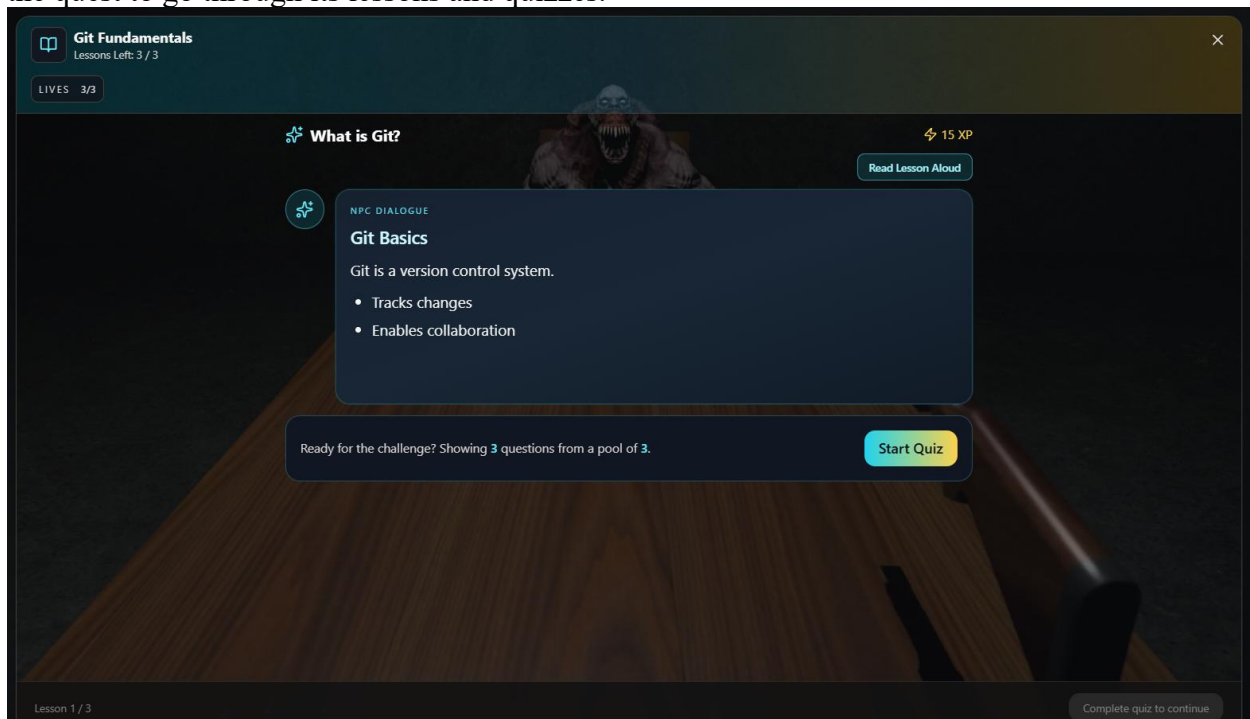


Figure 36: lesson player showing lesson content

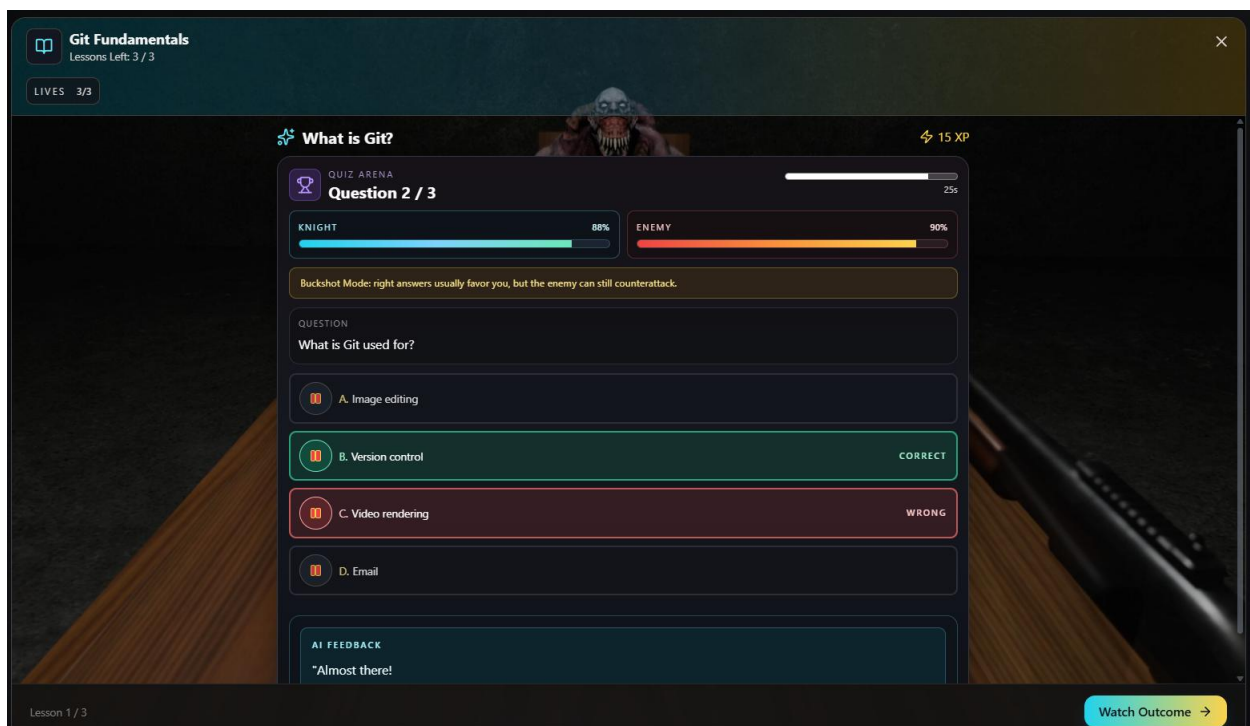


Figure 37: lesson player showing the quiz

After the submission of each answer AI feedback is received on the answer and a video is played showing the outcome of your answer.

User profile page

Here, the user can add images, display badges or see their journey on the site.



Figure 38: user profile

Settings

The settings page allows users to manage profile details like phone number, email address visibility and address.

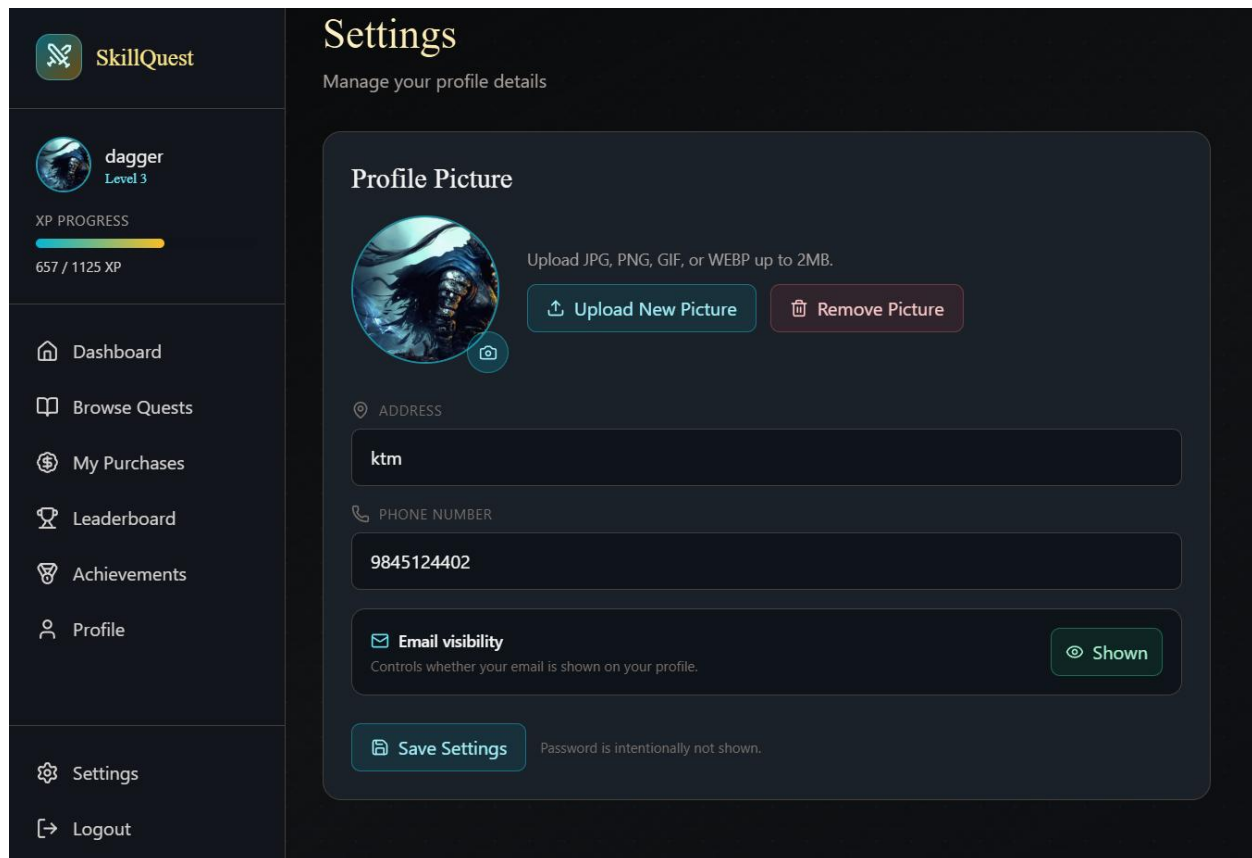


Figure 39: settings page

GitHub Commit History

Git and GitHub were used for version control throughout the development process. Regular commits were maintained to track feature implementation, bug fixes, UI improvements, and documentation updates.

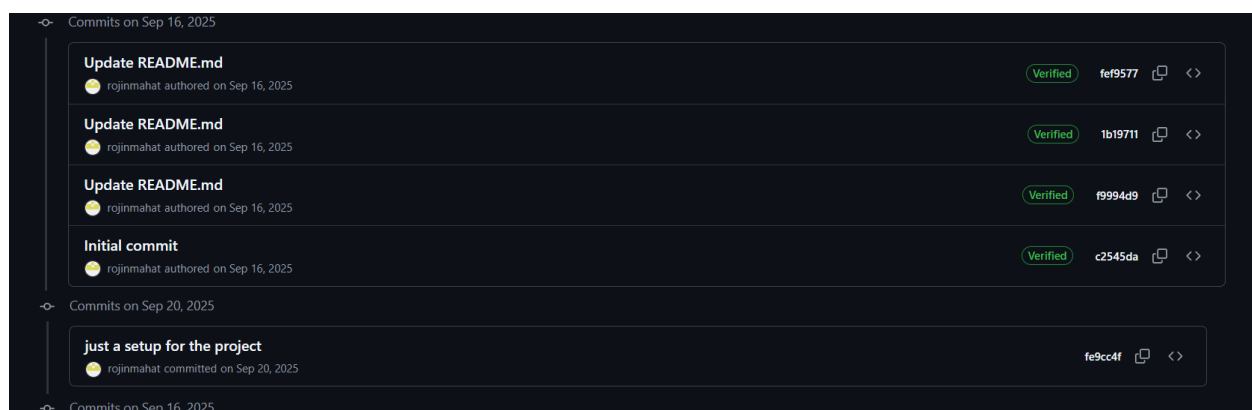


Figure 40: github first commit

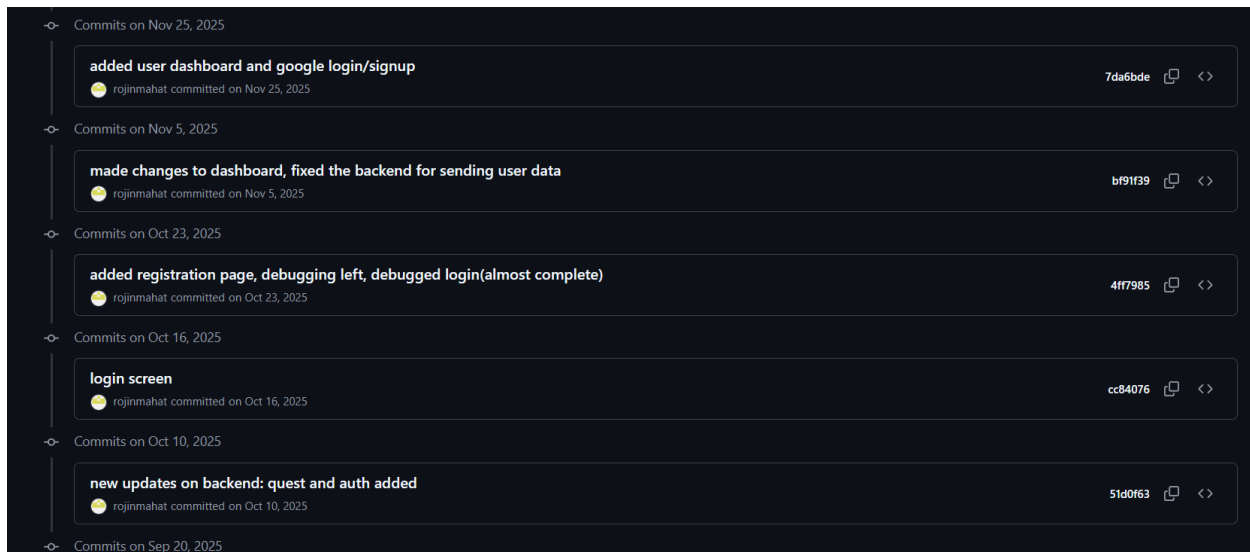


Figure 41: rest of github commits

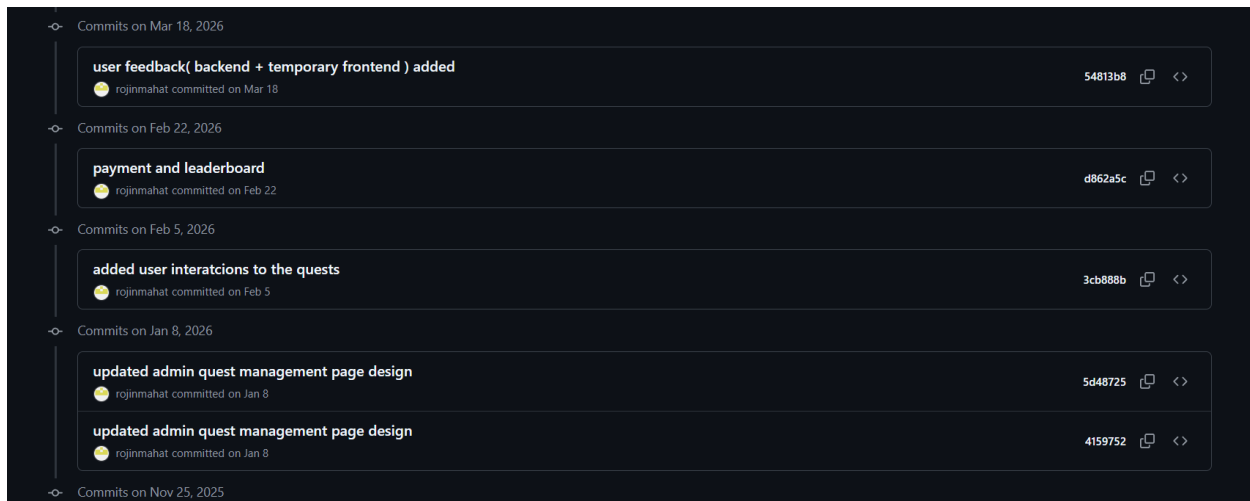


Figure 42: rest of github commits

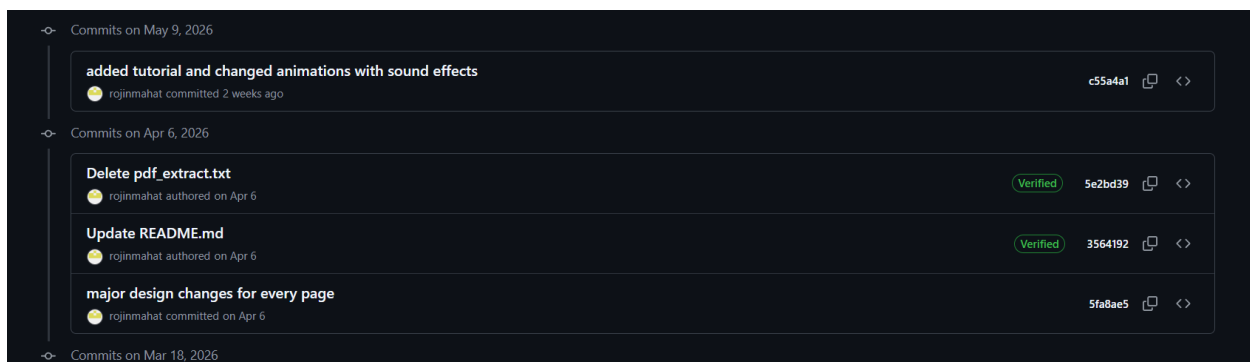


Figure 43: rest of github commits

PRF

Presented below are PRF documents signed by me and the Reader.

**Faculty of Science and Engineering
School of Mathematics and Computer
Science**

Will the information or artefacts resulting from your project be available externally to the University?	YES / NO If yes, please complete an <u>External Agreement Form</u>	
Will your project involve: (a) Human participants (b) Data about humans (c) Sensitive information	YES / NO If yes, please complete Page 2 of this form. Otherwise, please sign the top of Page 3, and if necessary pass it on to your supervisor.	
Human Participants	Yes	No
If your proposal involves healthy adult participants, does the project have characteristics that may be detrimental to their physical or mental wellbeing?		✓
Does the proposal involve vulnerable participants (for example, are they under 18 years of age, do they have a disability or are mentally unable to consent)?		✓
Privacy		
Does the proposal involve processing of genetic information or personal data (e.g. health, sexual lifestyle, ethnicity, political opinion, religious or philosophical conviction)		✓
Does the proposal involve tracking the location or observation of people without their knowledge?		✓
Research on Animals		
Does the proposal involve research with animals?		✓
Research Involving Developing Countries		
Is any of the research involving one of the <u>Least Developed Countries</u>?		✓
Dual Use		
Does the research have direct military applications?		✓
Does the research have the potential for terrorist abuse?		✓

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School of Mathematics and Computer
Science**



MCS COMMITTEE ON THE ETHICS OF RESEARCH

Registration of a Research Project

This form must be completed by the Researcher(s) and, in cases where the applicant is a student, the project Supervisor to determine if the project requires approval by the FSE Research Ethics Committee.

YOUR DETAILS

First Name: Rojin

Surname: Mahat

Student Number: 2417733

Supervisor: Saddam Khan

Course: ~~BSc (Hons.) IT~~ BSc (Hons.) Computer Science

Project Title: Gamified learning app for technical microskills

YOUR PROJECT

State in no more than 100 words the problem(s) your project is aiming to address

Traditional technical education platforms suffer from low engagement and high dropout rates due to lengthy courses and lack of motivation. This project develops a gamified platform for teaching technical microskills through bite-sized, engaging lessons that reward progress and maintain learner interest.

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Project Title:

Student Name: Rojin Singh Mahat Student Number: 0417733

Supervisor Name: Sadlam Khan

- We confirm that the information given in this form is true, complete and accurate.

Student Signature: Rojin Date: 2025/11/28

Supervisor Signature: Sadlam Date: 2025/11/28

Thank you for completing this form. The MCS Ethics Committee will process the information provided and inform you of their decision shortly.

Log sheets

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School of Mathematics and Computer Science



PROJECT MANAGEMENT LOG	
First Name: <u>Rojin</u>	Surname: <u>Mahol</u>
Student Number: <u>2417733</u>	Supervisor: <u>Saddam Khan</u>
Project Title: <u>SkillQuest (an interactive learning app)</u> Month: <u>November</u>	
What have you done since the last meeting	
The backend was successfully developed using MERN stack, featuring JWT-based authentication and MongoDB integration. User registration, login, and user-data retrieval end-point routes were implemented and tested. On the frontend, responsive login and registration interfaces were developed using React and tailwindcss, maintaining a modern and consistent design theme. Token handling and auth.-based redirection were integrated. Initial development of user dashboard was initiated. Also created MongoDB models, and a landing page	
What do you aim to complete before the next meeting	
Complete dashboard interface, connect frontend to backend APIs for data display, implement quest browsing page, create sample quest content.	
Supervisor comments	
> Frontend & backend part for user dashboard > And also work on artifacts design	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: Rojin

Date: 2025/11/16

Supervisor Signature: Saddam Khan

Date: 23/11/2025

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School of Mathematics and Computer Science



PROJECT MANAGEMENT LOG	
First Name: Rojin	Surname: Mahat
Student Number: 2417733	Supervisor: Saddam Khan
Project Title: Gamified learning app for micro-stills Month: November	
What have you done since the last meeting	
Added google auth Added frontend and backend for user dashboard Working on artifact design	
What do you aim to complete before the next meeting	
Improve ^{normal} authentication (Make sure email is legit and in-use) Add on quest browsing page admin dashboard (frontend & backend) Allow admins to CRUD quests and lessons Add a basic admin dashboard with working backend	
Supervisor comments	
> wants on dashboard mail factor	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: Rojin

Date: 2025/11/23

Supervisor Signature: Sakh

Date: 21/12/2025

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School of Mathematics and Computer Science

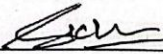


PROJECT MANAGEMENT LOG	
First Name: <u>Rojin</u>	Surname: <u>Mahat</u>
Student Number: <u>2417733</u>	Supervisor: <u>M.D. Saeldam Khan</u>
Project Title: <u>Hamified learning app for microskills</u> Month: <u>December</u>	
What have you done since the last meeting	
→ Added dashboard for all actors → Admins can now add, edit and delete quests	
What do you aim to complete before the next meeting	
→ Allow users to interact with the quests	
Supervisor comments	
> work on quest part for admin users so that they can read the document.	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 2025/12/21

Supervisor Signature: 

Date: 1-feb-2026

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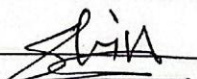


PROJECT MANAGEMENT LOG	
First Name: <u>Rojin</u>	Surname: <u>Mahat</u>
Student Number: <u>2457733</u>	Supervisor: <u>M.D. Saddam Khan</u>
Project Title: <u>Gamified Learning app</u>	Month: <u>January</u>
What have you done since the last meeting	
- Designed quest interaction logic and database structure	
What do you aim to complete before the next meeting	
- Implement Exp and badge reward system	
Supervisor comments	
> work on exp and badge systems and for	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 2026/01/04

Supervisor Signature: 

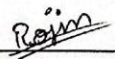
Date: 1-feb-2026

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School of Mathematics and Computer Science

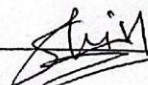


PROJECT MANAGEMENT LOG	
First Name: <u>Rojin</u>	Surname: <u>Mahad</u>
Student Number: <u>2617733</u>	Supervisor: <u>MD Saddam Khan</u>
Project Title: <u>Gamified learning app</u> Month: <u>January</u>	
What have you done since the last meeting	
<u>⇒ Integrated admin dashboard with quest management</u>	
What do you aim to complete before the next meeting	
<u>- Testing and UI improvements</u>	
Supervisor comments	
<u>> Find the necessary bugs and make improvements</u>	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 2026/01/18

Supervisor Signature: 

Date: 1-feb-2026

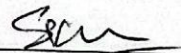
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PROJECT MANAGEMENT LOG	
First Name: <u>Rojin</u>	Surname: <u>Mahat</u>
Student Number: <u>2417733</u>	Supervisor: <u>M.D. Saddam Khan</u>
Project Title: <u>Gamified learning app</u>	Month: <u>February</u>
What have you done since the last meeting	
- Made small changes to design on admin page dashboard - Added Implemented user interaction with quests. i.e. Users can complete quests, gain exp and badges	
What do you aim to complete before the next meeting	
- Implement ^{payment} system for completing advanced and intermediate quests	
Supervisor comments	
> work on Payment Integration.	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 2026/02/01

Supervisor Signature: 

Date: 22/02/2026

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School of Mathematics and Computer Science

PROJECT MANAGEMENT LOG	
First Name: Rojin	Surname: Mhat
Student Number: 2417733	Supervisor: M.P Saddam Khan
Project Title: Gamified learning app for microskills Month: February	
What have you done since the last meeting	
→ Added payment method paid quests → Added payment method for paid quests (using e-sewa) → Fixed the bug where google sign up didn't send email to the user → Added a simple leaderboard	
What do you aim to complete before the next meeting	
→ Improve the frontend designs on figma → Add user feedback option for quests	
Supervisor comments	
→ Add user feedback option for quests	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: Rojin

Date: 2026/02/22

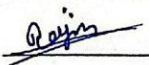
Supervisor Signature: [Signature]

Date: 8/03/2026

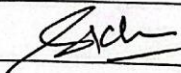
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School of Mathematics and Computer Science

PROJECT MANAGEMENT LOG	
First Name: <u>Rojin</u>	Surname: <u>Mahat</u>
Student Number: <u>2617733</u>	Supervisor: <u>MD Saddam Khan</u>
Project Title: <u>Skill Quest</u>	Month: <u>March</u>
What have you done since the last meeting	
- Added user feed back (Ratings, and admins, other learners can see the reviews) - Started work on wireframe/design on figma	
What do you aim to complete before the next meeting	
- <u>Design</u> and frontend for landing page and auth pages	
Supervisor comments	
> Design & frontend for landing page & Authentication.	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 2026/03/08

Supervisor Signature: 

Date: 15/03/2026

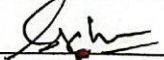
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School of Mathematics and Computer Science**

PROJECT MANAGEMENT LOG	
First Name: <u>Rojin</u>	Surname: <u>Mahat</u>
Student Number: <u>2417783</u>	Supervisor: <u>MD Saddam Khan</u>
Project Title: <u>Swill quest</u>	Month: <u>March</u>
What have you done since the last meeting	
- Designed landing page, login page, Signup page - Made frontend for designed pages	
What do you aim to complete before the next meeting	
- Designe Dashboard, leaderboard, Browse quest, profile pages - Make frontend for designed pages	
Supervisor comments	
> work on dashboard, leaderboard, profile pages.	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 2026/03/105

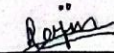
Supervisor Signature: 

Date: 12/04/2026

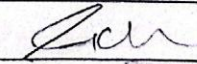
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PROJECT MANAGEMENT LOG	
First Name: <u>Rojin Singh Mahat</u>	Surname: <u>Mahat</u>
Student Number: <u>2617733</u>	Supervisor: <u>MD Saddam Khan</u>
Project Title: <u>Skill quest</u>	Month: <u>April</u>
What have you done since the last meeting	
- Changed designs for all the pages. - Added achievements - Added Profile management - Bug fixes: email and password validation during sign up - changed	
What do you aim to complete before the next meeting	
- Add game animations to lessons	
Supervisor comments	
> task Complete dashboard work. and animation part	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 2026/04/12

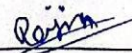
Supervisor Signature: 

Date: 4/29/2026

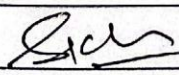
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School of Mathematics and Computer Science

PROJECT MANAGEMENT LOG	
First Name: <u>Rojin</u>	Surname: <u>Mahat</u>
Student Number: <u>2417733</u>	Supervisor: <u>MD Saddam Khan</u>
Project Title: <u>Skill Quest</u>	Month: <u>April</u>
What have you done since the last meeting	
- Made Animations for the lesson player - Added said animations to the lesson player	
What do you aim to complete before the next meeting	
- Add a tutorial mode for new users - Add more animations for other lesson	
Supervisor comments	
> Start working on bug fixed testing	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 2026/04/19

Supervisor Signature: 

Date: 3/05/2026

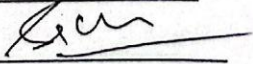
**Faculty of Science and Engineering
School of Mathematics and Computer Science**

PROJECT MANAGEMENT LOG	
First Name: <u>Rojin</u>	Surname: <u>Mahat</u>
Student Number: <u>2477233</u>	Supervisor: <u>115 Saddam Khar</u>
Project Title: <u>SkillsQuest</u>	Month: <u>April May</u>
What have you done since the last meeting	
- Added user tutorial - changed game animation	
What do you aim to complete before the next meeting	
- Add 25 feedbacks for each quiz - Show all analytics	
Supervisor comments	
> allocate an analytics part	

We confirm that the information given in this form is true, complete and accurate.

Student Signature: 

Date: 2026/05/30

Supervisor Signature: 

Date: 10/05/2026